STAT8561 - Linear Statistical Analysis I

2026 Fall Lecture Notes

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Preface

The topic of this course includes statistical inference, Multivariate normal distribution, distribution of quadratic forms, linear models, regression models and experimental design models.

Prerequisites

Math 4751/6751 Mathematical Statistics I and Math 4752/6752 Mathematics and Statistics II.

Instructor

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Office Hour

9:00 - 12:00 on Wednesday or by appointment

Grade Distribution

- Homework – 40%
- Exam 1 – 20%
- Exam 2 – 20%
- Final – 20%
- Attendance – *7%

Assignment

☒ A1, Due September 11, 2026
☐ A2, TBA

Exam

☐ Exam 1, October 14, 2026

Topics and Corresponding Lectures

Those chapters are based on the lecture notes. This part will be updated frequently.

Status	Chapter	Topic	Lecture
		Welcome and Overview	1
	Ch. 1	Math Notation	2–3
	Ch. 2	Least Squares Estimation	3–
	Ch. 3	Inference on Least Square Estimation	3–

Recommended Textbooks

- Ding, Peng. *Linear Model and Extensions.* CRC Press, 2027. arXiv link
- Montgomery, Douglas C., Elizabeth A. Peck, and G. Geoffrey Vining. *Introduction to Linear Regression Analysis.* John Wiley & Sons, 2021. (Montgomery et al. 2021)
- Seber, George AF, and Alan J. Lee. *Linear Regression Analysis.* John Wiley & Sons, 2003. (Seber and Lee 2003)

Side Readings

- Montgomery, Douglas C. (2017). *Design and Analysis of Experiments.* John Wiley & Sons. (Montgomery 2017)

Introduction

This is an introduction to the course for linear statistical analysis. It will give you an overview of what we will be covering in the course and how to get the most out of it. We will be covering a wide range of topics in linear statistical analysis,

- simple linear regression
- multiple linear regression
- generalized linear models
- mixed effects models
- quantile regression.

We will also be discussing the assumptions underlying these models and how to check them.

The course will be structured around lectures, homework assignments, a midterm and a final.

To get the most out of this course, it is important to attend all lectures and complete all homework assignments. It is also important to ask questions and participate in class discussions. The more you engage with the material, the more you will learn.

I am looking forward to a great semester and I hope you are too!

To illustrate the concepts we will be covering in this course, let's consider a simple example. Suppose we have a data set of heights and weights of individuals. We want to understand the relationship between height and weight, and we can use linear regression to model this relationship.

```
# Load necessary libraries
library(ggplot2)
library(dplyr)
# Create a sample data set

set.seed(8561)

theme_set(theme_minimal())

n <- 100
heights <- rnorm(n, mean = 170, sd = 10)
weights <- 0.5 * heights + rnorm(n, mean = 0, sd
```

```r
 = 5)
df <- data.frame(heights, weights)
head(df, 5)
```

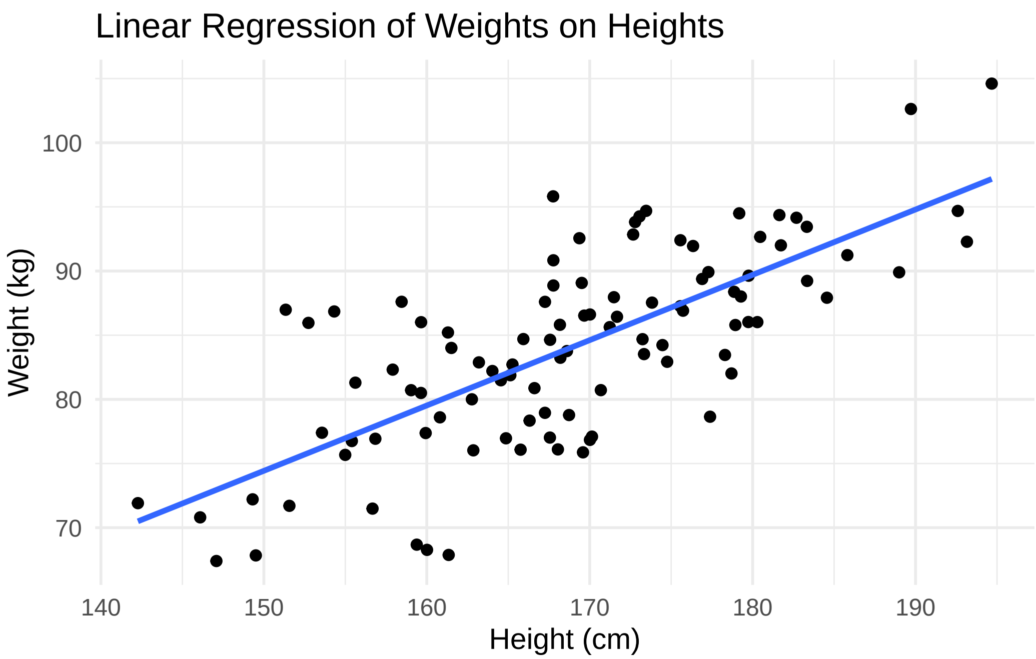
```
   heights  weights
1 179.2812 88.01599
2 168.0474 76.10017
3 178.3029 83.46016
4 178.7006 82.01763
5 172.6678 92.84592
```

```r
# Fit a linear regression model
model <- lm(weights ~ heights, data = df)
# Summarize the model
summary(model)
```

```

Call:
lm(formula = weights ~ heights, data = df)

Residuals:
     Min       1Q   Median       3Q      Max 
-12.3303  -3.8998  -0.2191   3.2380  12.3478 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.92852    8.22226  -0.235    0.815    
heights      0.50907    0.04857  10.481   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.251 on 98 degrees of freedom
Multiple R-squared:  0.5285,	Adjusted R-squared:  0.5237 
F-statistic: 109.9 on 1 and 98 DF,  p-value: < 2.2e-16
```

```r
# Plot the data and the fitted line
ggplot(df, aes(x = heights, y = weights)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Linear Regression of Weights on Heights",
       x = "Height (cm)",
       y = "Weight (kg)")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



In this example, we generated a data set of heights and weights, fitted a linear regression model to the data, and visualized the relationship between height and weight. This is just a simple example, but it illustrates the types of analyses we will be doing in this course. We will be covering much more complex models and data sets as we progress through the semester.

Throughout the course, we will be using **R** programming software.

1 Matrix Representation and Notation

In this chapter, we introduce the mathematical language and statistical framework that will be used throughout the course. Our focus is on the notation of vectors and matrices, vectors of random variables, expectation and covariance operators, and the basic form of the linear regression model.

Learning Objectives

By the end of this chapter, students should be able to:

- use standard matrix notation for linear statistical models;
- distinguish between scalars, vectors, matrices, random variables, and random vectors;
- compute expectations and covariance matrices for random vectors;
- interpret the linear regression model in matrix form;
- understand why projection and least squares will play a central role in this course.

Reading

Recommended reading for this chapter:

- Seber and Lee (2003), Chapter 1:
  - 1.1 Notation
  - 1.2 Statistical Models
  - 1.3 Linear Regression Models
  - 1.4 Expectation and Covariance Operators
- Optional preview:
  - Chapter 2: Multivariate Normal Distribution

1.1 Why Linear Statistical Analysis?

Linear statistical analysis is one of the central foundations of graduate statistics. Many methods that at first look different are built on the same underlying structure:

- regression
- analysis of variance
- analysis of covariance
- prediction
- model comparison
- and parts of generalized linear modelling

A major goal of this course is to see these topics under a unified framework.

1.2 A unifying point of view

A large part of the course can be summarized by the model

$$Y = X\beta + \varepsilon,$$

where

- Y is a response vector,
- X is a design matrix,
- β is an unknown parameter vector,
- ε is a random error vector.

This compact expression contains a great deal of statistical structure. Over the semester, we will study how to estimate β, quantify uncertainty, test hypotheses, diagnose model failures, and make predictions.

1.3 Basic Notation

Scalars, vectors, and matrices

We use the following conventions throughout the course:

- scalars are written in lowercase italic letters, such as a, b, n;
- vectors are written in bold lowercase letters, such as $\mathbf{x}$, $\mathbf{y}$;
- matrices are written in bold uppercase letters, such as $\mathbf{X}$, $\mathbf{A}$;
- random variables are often written in uppercase letters, such as Y;
- realizations of random variables are written in lowercase letters, such as y.

Univariate case

A random variable is **univariate** if it takes values in one dimension. We write

$$X \in \mathbb{R}.$$

A realization, or observed value, of X is denoted by x.

Some common quantities associated with a univariate random variable X include:

- **Mean**
$$\mu = \mathbb{E}(X).$$
- **Variance**
$$\sigma^2 = \mathrm{Var}(X) = \mathbb{E}\left[(X-\mu)^2\right].$$
- **kth moment**
$$\mathbb{E}(X^k).$$
- **Cumulative distribution function (CDF)**
$$F(x) = P(X \leq x).$$
- **Probability density function (PDF)**, for a continuous random variable,
$$f(x).$$
- **Probability mass function (PMF)**, for a discrete random variable,
$$p(x) = P(X = x).$$
- **Quantile**
$$q_\alpha = F^{-1}(\alpha),$$
where $0 < \alpha < 1$.

For a random sample of size n, we write

$$X_1, X_2, \dots, X_n,$$

with observed values

$$x_1, x_2, \dots, x_n.$$

The sample mean is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i,$$

and the sample variance is

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Multivariate case

A column vector in $\mathbb{R}^n$ is written as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = (x_1, x_2, \ldots, x_n).$$

An $n \times p$ matrix is written as

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}.$$

Transpose, inverse, and rank

If $\mathbf{A}$ is a matrix, then:

- $\mathbf{A}^\top$ denotes its transpose;
- $\mathbf{A}^{-1}$ denotes its inverse, when it exists;
- $\operatorname{rank}(\mathbf{A})$ denotes its rank;
- $\mathbf{I}_n$ denotes the $n \times n$ identity matrix.

1.3.1 Inner product and norm

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the inner product is

$$\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^{n} x_i y_i.$$

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the inner product is

$$\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}.$$

These ideas are fundamental because least squares estimation is based on minimizing squared Euclidean distance.

A random vector is a vector whose entries are random variables. For example,

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}$$

is an n-dimensional random vector.

In linear models, the response is naturally treated as a random vector.

The mean vector of $\mathbf{Y}$ is

$$\mathbb{E}[\mathbf{Y}] = \begin{pmatrix} \mathbb{E}[Y_1] \\ \mathbb{E}[Y_2] \\ \vdots \\ \mathbb{E}[Y_n] \end{pmatrix}.$$

We often write $\mathbb{E}[\mathbf{Y}] = \mu$.

The covariance matrix of $\mathbf{Y}$ is

$$\mathrm{Var}(\mathbf{Y}) = \mathbb{E}\left[(\mathbf{Y} - \mu)(\mathbf{Y} - \mu)^\top\right].$$

Equivalently,

$$\mathrm{Var}(\mathbf{Y}) = \mathbb{E}[\mathbf{Y}\mathbf{Y}^\top] - \mu\mu^\top.$$

Suppose $\mathbf{Y} = (Y_1, Y_2, Y_3)$.

Then the corresponding covariance matrix is

$$\mathrm{Var}(\mathbf{Y}) = \begin{pmatrix} \mathrm{Var}(Y_1) & \mathrm{Cov}(Y_1, Y_2) & \mathrm{Cov}(Y_1, Y_3) \\ \mathrm{Cov}(Y_2, Y_1) & \mathrm{Var}(Y_2) & \mathrm{Cov}(Y_2, Y_3) \\ \mathrm{Cov}(Y_3, Y_1) & \mathrm{Cov}(Y_3, Y_2) & \mathrm{Var}(Y_3) \end{pmatrix}.$$

1.3.2 Expectation and Covariance Operators

If $\mathbf{A}$ is a constant matrix and $\mathbf{b}$ is a constant vector, then

$$\mathbb{E}[\mathbf{A}\mathbf{Y} + \mathbf{b}] = \mathbf{A}\mathbb{E}[\mathbf{Y}] + \mathbf{b}.$$

This is one of the most useful identities in the course.

If $\mathbf{A}$ is a constant matrix, then

$$\mathrm{Var}(\mathbf{A}\mathbf{Y}) = \mathbf{A}\,\mathrm{Var}(\mathbf{Y})\,\mathbf{A}^\top.$$

More generally, if $\mathbf{Y}$ and $\mathbf{Z}$ are random vectors, then

$$\mathrm{Cov}(\mathbf{A}\mathbf{Y}, \mathbf{B}\mathbf{Z}) = \mathbf{A}\,\mathrm{Cov}(\mathbf{Y}, \mathbf{Z})\,\mathbf{B}^\top.$$

These formulas are essential for deriving the variance of estimators later.

If $Y_1, \dots, Y_n$ are independent and each has variance σ^2, then

$$\mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

This is the most common starting assumption in classical linear regression.

1.4 Statistical Models

A statistical model is a set of probability distributions that may plausibly describe the data-generating mechanism.

1.4.1 General idea

Suppose we observe data y from a random quantity Y. A model introduces assumptions about the distribution of Y, often indexed by an unknown parameter θ.

For example:

$$Y \sim N(\mu, \sigma^2)$$

with unknown parameters μ and σ^2.

In regression, the model is not only about the marginal distribution of the response but also about how the mean changes with explanatory variables.

1.4.2 Deterministic part and random part

A useful way to think about a statistical model is:

$$\text{data} = \text{systematic part} + \text{random part}.$$

For linear regression, this becomes

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

Here,

- $\beta_0 + \beta_1 x_i$ is the systematic part;
- ε_i is the random part.

1.5 The Linear Regression Model

Perhaps the most simple model is the *simple linear regression* where there are only **one** independent variable and **one** response variable.

A simple linear regression model is defined as

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \dots, n,$$

with the typical assumption as

$$\mathbb{E}[\varepsilon_i] = 0, \qquad \mathrm{Var}(\varepsilon_i) = \sigma^2, \qquad \mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0 \text{ for } i \neq j.$$

Matrix form

We can write the model compactly as

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon,$$

where

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \qquad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \qquad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

This notation will allow us to treat simple regression, multiple regression, ANOVA, and ANCOVA in one common language.

Under the same assumption,

$$\mathbb{E}[\varepsilon] = \mathbf{0} \qquad \text{and} \qquad \mathrm{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n,$$

we have the following.

$$\mathbb{E}[\mathbf{Y}] = \mathbf{X}\beta, \quad \mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

These are immediate consequences of the expectation and covariance rules above.

1.5.1 Geometry Preview

A central idea in linear regression is projection.

The fitted values $\hat{\mathbf{Y}}$ will later be obtained by projecting $\mathbf{Y}$ onto the column space of $\mathbf{X}$.

The column space of $\mathbf{X}$ is

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\beta : \beta \in \mathbb{R}^p\}.$$

This is the set of all mean vectors that the model can represent.

1.6 Why projection matters

The least squares estimator chooses $\widehat{\beta}$ so that

$$\|\mathbf{Y} - \mathbf{X}\beta\|^2$$

is minimized.

So regression is not only algebra. It is also geometry.

Suppose we observe the following data:

x_i	0	1	2	3
y_i	1	3	3	5

Then

$$\mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}.$$

We will learn next chapter that the least squares estimator is

$$\widehat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

For now, the main point is to understand how the model is written and how the data are represented in matrix form.

R code

```
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

mod <- lm(y ~ x)
summary(mod)
```

```
Call:
lm(formula = y ~ x)

Residuals:
   1    2    3    4 
-0.2  0.6 -0.6  0.2 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)  
(Intercept)   1.2000     0.5292   2.268   0.1515  
x             1.2000     0.2828   4.243   0.0513 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6325 on 2 degrees of freedom
Multiple R-squared:    0.9,	Adjusted R-squared:   0.85 
F-statistic:    18 on 1 and 2 DF,  p-value: 0.05132
```

```r
# Inspecting the design matrix
model.matrix(mod)
```

```
  (Intercept) x
1           1 0
2           1 1
3           1 2
4           1 3
attr(,"assign")
[1] 0 1
```

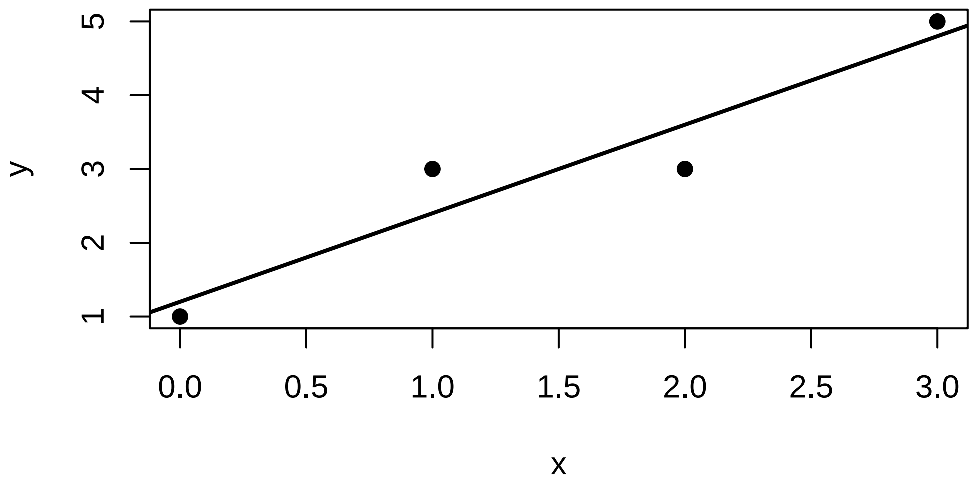
```r
# Fitted values and residuals
fitted(mod)
```

```
  1   2   3   4 
1.2 2.4 3.6 4.8 
```

```r
residuals(mod)
```

```
   1    2    3    4 
-0.2  0.6 -0.6  0.2 
```

```r
# quick plot
plot(x, y, pch = 19, xlab = "x", ylab = "y")
abline(mod, lwd = 2)
```



1.7 Discussion

1. Why is it helpful to write regression models in matrix form rather than only scalar notation?
2. What does the covariance matrix tell us that separate variances do not?
3. What is the interpretation of the column space of $\mathbf{X}$?
4. In what sense is regression a projection problem?

1.8 Practice Problems

Conceptual

1. Explain the difference between a random variable and a random vector.
2. Explain why $\mathrm{Var}(\mathbf{Y})$ must be a symmetric matrix.

3. Give an example of a statistical model outside regression.

Computational

Let

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}$$

with

$$\mathbb{E}[\mathbf{Y}] = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad \mathrm{Var}(\mathbf{Y}) = \begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}.$$

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

Compute:

1. $\mathbb{E}[\mathbf{AY} + \mathbf{b}]$
2. $\mathrm{Var}(\mathbf{AY})$

Regression setup

For the model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

write out the matrices $\mathbf{Y}$, $\mathbf{X}$, β, and ε for $n = 5$ observations.

1.9 Take home Practice

Complete the following:

- review matrix multiplication and transpose rules;
- derive $\mathbb{E}[\mathbf{AY} + \mathbf{b}]$ from first principles;
- derive $\mathrm{Var}(\mathbf{AY})$ using the definition of covariance;
- write the simple linear regression model in matrix form for a dataset of your choice;
- fit a simple regression in R and report:
- the estimated coefficients,

- fitted values,
- residuals,
- and a scatterplot with the fitted line.

1.10 Summary

This chapter introduced the notation and basic probabilistic tools needed for the rest of the course. We defined random vectors, mean vectors, covariance matrices, and the matrix form of the linear regression model. These ideas will support everything that follows.

Next chapter, we will study least squares estimation and the geometry of projection in more detail.

For some optional review of the matrix algebra, see Matrix Algebra Review.

2 Least Squares Estimation

In this chapter, we study one of the fundamental ideas in linear models: **least squares estimation**.

Recall that the linear model is

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon.$$

The unknown parameter vector β determines the mean structure of the response. Given observed data, our first goal is to estimate β.

The central idea of least squares is simple:

> Choose the value of β for which the fitted response $\mathbf{X}\beta$ is as close as possible to the observed response $\mathbf{Y}$.

This optimization problem leads to the **normal equations**, the **least squares estimator**, and a useful geometric interpretation in terms of **orthogonal projection**.

> Learning Objectives
>
> By the end of this chapter, students should be able to:
>
> - formulate least squares estimation as an optimization problem;
> - define the residual sum of squares;
> - derive the normal equations;
> - obtain the least squares estimator when $\mathbf{X}$ has full column rank;
> - explain the role of the column space $\mathcal{C}(\mathbf{X})$;
> - interpret fitted values as an orthogonal projection;
> - explain why residuals are orthogonal to the model space;
> - define and interpret the hat matrix and residual-maker matrix;
> - verify symmetry and idempotence of projection matrices;
> - explain the decomposition $\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$;
> - distinguish the uniqueness of $\hat{\beta}$ from the uniqueness of the fitted values.

Reading

Recommended reading for this week:

- Seber and Lee:
  - Chapter 3: sections on least squares estimation
- Montgomery, Peck, and Vining:
  - introductory sections on estimation in linear regression

Recap of the Linear Model

Recall from Week 1, the linear model can be written as

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon,$$

where

- $\mathbf{Y}$ is an $n \times 1$ response vector;
- $\mathbf{X}$ is an $n \times p$ design matrix;
- β is a $p \times 1$ parameter vector;
- ε is an $n \times 1$ error vector.

The design matrix may be written as

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix},$$

where $\mathbf{x}_i \in \mathbb{R}^p$ contains the predictors associated with observation i.
Thus,

$$Y_i = \mathbf{x}_i^\top \beta + \varepsilon_i, \qquad i = 1, \dots, n.$$

Under the classical linear model, we often assume

$$\mathbb{E}(\varepsilon) = \mathbf{0}, \qquad \mathrm{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n.$$

Consequently,

$$\mathbb{E}(\mathbf{Y}) = \mathbf{X}\beta, \qquad \mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

Least Squares Does Not Require a Distributional Assumption

The least squares estimator is defined through the optimization problem

$$\min_{\beta} \|\mathbf{Y} - \mathbf{X}\beta\|_2^2 .$$

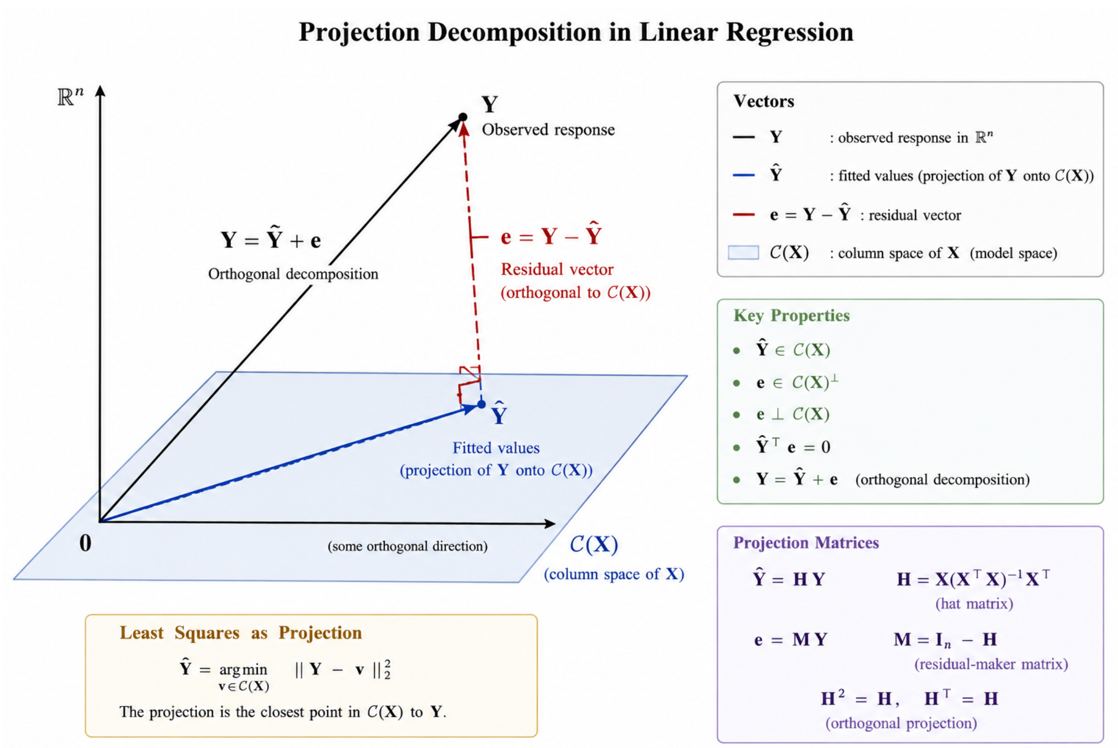


Figure 2.1: Geometric interpretation of least squares as an orthogonal projection.

No distributional assumption on ε is required to define or compute the least squares estimator.
Assumptions such as

$$\mathbb{E}(\varepsilon) = \mathbf{0}$$

and

$$\mathrm{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n$$

are needed when we study the **statistical properties** of the estimator.
Normality will be introduced later when we study exact finite-sample inference.

2.1 The Least Squares Criterion

Motivation

For any candidate value β, the corresponding fitted vector is

$$\mathbf{X}\beta.$$

The discrepancy between the observed response $\mathbf{Y}$ and this candidate fit is

$$\mathbf{Y} - \mathbf{X}\beta.$$

A natural idea is to choose β so that this discrepancy is as small as possible.

To measure its size, we use the squared Euclidean norm.

Residual Sum of Squares

The **residual sum of squares** associated with a candidate β is

$$S(\beta) = \|\mathbf{Y} - \mathbf{X}\beta\|_2^2 .$$

Equivalently,

$$S(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta).$$

In scalar notation,

$$S(\beta) = \sum_{i=1}^{n} \left(Y_i - \mathbf{x}_i^\top \beta\right)^2.$$

This last expression explains the name **least squares**: we choose β to make the sum of the squared discrepancies as small as possible.

We therefore define

$$\hat{\beta} = \arg\min_{\beta \in \mathbb{R}^p} S(\beta).$$

Definition 2.1 (Least Squares Estimator)**.** A **least squares estimator** is any value $\hat{\beta}$ satisfying

$$S(\hat{\beta}) \leq S(\beta)$$

for every $\beta \in \mathbb{R}^p$.

Example 2.1 (LSE for Simple Linear Regression)**.** Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \dots, n,$$

where

$$\mathbb{E}[\varepsilon_i] = 0, \qquad \mathrm{Var}(\varepsilon_i) = \sigma^2.$$

The fitted value at x_i is

$$\hat{Y}_i = \beta_0 + \beta_1 x_i.$$

The corresponding residual is

$$e_i = Y_i - \beta_0 - \beta_1 x_i.$$

The least squares method chooses β_0 and β_1 to minimize the **sum of squared residuals**

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} \left(Y_i - \beta_0 - \beta_1 x_i\right)^2.$$

Derivation of the Normal Equations

To minimize $S(\beta_0, \beta_1)$, differentiate with respect to β_0 and β_1.

For β_0,

$$\frac{\partial S}{\partial \beta_0} = -2 \sum_{i=1}^{n} \left(Y_i - \beta_0 - \beta_1 x_i\right).$$

Setting this equal to zero gives

$$\sum_{i=1}^{n} \left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right) = 0.$$

Therefore,

$$\sum_{i=1}^{n} Y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} x_i.$$

Dividing by n,

$$\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x},$$

so

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}}.$$

For β_1,

$$\frac{\partial S}{\partial \beta_1} = -2 \sum_{i=1}^{n} x_i \left(Y_i - \beta_0 - \beta_1 x_i\right).$$

Setting this equal to zero gives

$$\sum_{i=1}^{n} x_i \left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right) = 0.$$

Hence,

$$\sum_{i=1}^{n} x_i Y_i = \hat{\beta}_0 \sum_{i=1}^{n} x_i + \hat{\beta}_1 \sum_{i=1}^{n} x_i^2.$$

Substituting

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x},$$

we obtain

$$\sum_{i=1}^{n} x_i Y_i = \left(\bar{Y} - \hat{\beta}_1 \bar{x}\right) \sum_{i=1}^{n} x_i + \hat{\beta}_1 \sum_{i=1}^{n} x_i^2.$$

Since

$$\sum_{i=1}^{n} x_i = n\bar{x},$$

we have

$$\sum_{i=1}^{n} x_i Y_i - n\bar{x}\bar{Y} = \hat{\beta}_1 \left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2 \right).$$

Using

$$\sum_{i=1}^{n} (x_i - \bar{x})(Y_i - \bar{Y}) = \sum_{i=1}^{n} x_i Y_i - n\bar{x}\bar{Y},$$

and

$$\sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2,$$

we obtain

$$\boxed{\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}}.$$

Therefore,

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}}.$$

Alternative Notation

It is common to define

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

and

$$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(Y_i - \bar{Y}).$$

Then

$$\boxed{\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}}$$

and

$$\boxed{\hat{\beta}_0 = \bar{Y} - \frac{S_{xy}}{S_{xx}} \bar{x}}.$$

Thus, the fitted regression line is

$$\boxed{\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i}.$$

2.2 Derivation of the Least Squares Estimator in a matrix form

Expansion of the Criterion Starting from

$$S(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta),$$

we expand:

$$S(\beta) = \mathbf{Y}^\top \mathbf{Y} - \mathbf{Y}^\top \mathbf{X}\beta - \beta^\top \mathbf{X}^\top \mathbf{Y} + \beta^\top \mathbf{X}^\top \mathbf{X}\beta.$$

Because

$$\mathbf{Y}^\top \mathbf{X}\beta$$

is a scalar,

$$\mathbf{Y}^\top \mathbf{X}\beta = \beta^\top \mathbf{X}^\top \mathbf{Y}.$$

Therefore,

$$\boxed{S(\beta) = \mathbf{Y}^\top \mathbf{Y} - 2\beta^\top \mathbf{X}^\top \mathbf{Y} + \beta^\top \mathbf{X}^\top \mathbf{X}\beta.}$$

Derivation of the Normal Equations

Differentiate $S(\beta)$ with respect to β:

$$\frac{\partial S(\beta)}{\partial \beta} = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X}\beta.$$

At a minimum, the gradient is zero, so

$$-2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X}\hat{\beta} = \mathbf{0}.$$

Thus,

$$\boxed{\mathbf{X}^\top \mathbf{X}\hat{\beta} = \mathbf{X}^\top \mathbf{Y}.}$$

These are called the **normal equations**.

Why Are They Called the Normal Equations?

The normal equations can be written as

$$\mathbf{X}^\top \left(\mathbf{Y} - \mathbf{X}\hat{\beta}\right) = \mathbf{0}.$$

Later we will define

$$\mathbf{e} = \mathbf{Y} - \mathbf{X}\hat{\beta}.$$

Therefore,

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

This means that $\mathbf{e}$ is perpendicular, or **normal**, to the column space of $\mathbf{X}$. Thus, the word *normal* is being used in its geometric sense.

Why is the Stationary Point a Minimum?

The Hessian of the least squares criterion is

$$\nabla^2 S(\beta) = 2\mathbf{X}^\top \mathbf{X}.$$

For any $\mathbf{a} \in \mathbb{R}^p$,

$$\mathbf{a}^\top \mathbf{X}^\top \mathbf{X}\mathbf{a} = (\mathbf{X}\mathbf{a})^\top (\mathbf{X}\mathbf{a}) = \|\mathbf{X}\mathbf{a}\|_2^2 \geq 0.$$

Therefore, $\mathbf{X}^\top \mathbf{X}$ is positive semidefinite, and $S(\beta)$ is a convex function.

If $\mathbf{X}$ has full column rank, then for every $\mathbf{a} \neq \mathbf{0}$,

$$\mathbf{X}\mathbf{a} \neq \mathbf{0},$$

and hence

$$\mathbf{a}^\top \mathbf{X}^\top \mathbf{X}\mathbf{a} > 0.$$

Thus, $\mathbf{X}^\top \mathbf{X}$ is positive definite and the least squares minimizer is unique.

A vector $\widehat{\beta}$ minimizes

$$S(\beta) = \|\mathbf{Y} - \mathbf{X}\beta\|_2^2$$

if and only if it satisfies the normal equations

$$\mathbf{X}^\top \mathbf{X}\widehat{\beta} = \mathbf{X}^\top \mathbf{Y}.$$

If $\mathbf{X}$ has full column rank,

$$\operatorname{rank}(\mathbf{X}) = p,$$

then $\mathbf{X}^\top \mathbf{X}$ is invertible and the least squares estimator is unique:

$$\boxed{\widehat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.}$$

This is commonly called the **ordinary least squares estimator**, or **OLS estimator**.

When Is the Closed-Form Formula Valid?

The expression

$$(\mathbf{X}^\top \mathbf{X})^{-1}$$

exists if and only if $\mathbf{X}$ has full column rank:

$$\operatorname{rank}(\mathbf{X}) = p.$$

This means that the columns of $\mathbf{X}$ are linearly independent.
If they are linearly dependent, then $\mathbf{X}^\top \mathbf{X}$ is singular and the inverse does not exist.
We will return to this issue later.

Example 2.2 (LSE for Simple Linear Regression continue)**.** As shown in Example 2.1, we know the expression of the LSE$\hat{}$. Here, we want to see its connection to the matrix Form.

For simple linear regression,

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \qquad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}.$$

The general least squares estimator is

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

For simple linear regression,

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{pmatrix},$$

and

$$\mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} \sum_{i=1}^n Y_i \\ \sum_{i=1}^n x_i Y_i \end{pmatrix}.$$

Therefore,

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y},$$

which gives exactly

$$\boxed{\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

and

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}}.$$

Hence,

> The familiar simple linear regression formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$ are special cases of the general matrix LSE
>
> $$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

2.3 Geometric Interpretation of Least Squares

The algebraic derivation gives us the estimator. However, the geometry explains **why** least squares has its important properties.

> **Recap: Column Space of the Design Matrix**
>
> The column space of $\mathbf{X}$ is
>
> $$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\beta : \beta \in \mathbb{R}^p\}.$$
>
> Thus, $\mathcal{C}(\mathbf{X})$ is the set of all response vectors that can be represented exactly by the linear model.
> If $\mathbf{X}$ has full column rank p, then
>
> $$\dim \{\mathcal{C}(\mathbf{X})\} = p.$$
>
> Because $\mathbf{X}$ has n rows,
>
> $$\mathcal{C}(\mathbf{X}) \subseteq \mathbb{R}^n.$$

2.3.1 Least Squares as Projection

The fitted value vector is

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}.$$

Because

$$\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X}),$$

least squares chooses the vector in $\mathcal{C}(\mathbf{X})$ that is closest to the observed vector $\mathbf{Y}$.

That is,

$$\hat{\mathbf{Y}} = \arg \min_{\mathbf{v} \in \mathcal{C}(\mathbf{X})} \|\mathbf{Y} - \mathbf{v}\|_2^2.$$

Therefore, $\hat{\mathbf{Y}}$ is the **orthogonal projection** of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$.

> So we can say that the fitted vector $\hat{\mathbf{Y}}$ is constrained to live in the model space generated by the predictors. Least squares finds the point in that space closest to the observed data vector $\mathbf{Y}$.

Definition 2.2 (Orthogonality). Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are said to be **orthogonal** if their inner product is zero:

$$\mathbf{u}^\top \mathbf{v} = 0.$$

We write

$$\mathbf{u} \perp \mathbf{v}.$$

Geometrically, two nonzero vectors are orthogonal if the angle between them is 90°.

Definition 2.3 (Residual Vector). After obtaining the LSE, define the **residual vector**

$$\boxed{\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X}\hat{\beta}.}$$

The individual residuals are

$$e_i = Y_i - \widehat{Y}_i, \qquad i = 1, \dots, n.$$

Geometrically, $\mathbf{e}$ is the component of $\mathbf{Y}$ that is orthogonal to the model space $\mathcal{C}(\mathbf{X})$.

Proposition 2.1 (Orthogonality of the Residuals)**.** *Recall that the LSE satisfies the normal equations* $\mathbf{X}^\top \mathbf{X}\widehat{\beta} = \mathbf{X}^\top \mathbf{Y}$. *If we rearrange, we have* $\mathbf{X}^\top \left(\mathbf{Y} - \mathbf{X}\widehat{\beta}\right) = \mathbf{0}$.

Since the residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X}\widehat{\beta},$$

we obtain the fundamental least squares property

$$\boxed{\mathbf{X}^\top \mathbf{e} = \mathbf{0}.}$$

To understand what this means, write the design matrix as

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}^{(1)} & \mathbf{x}^{(2)} & \cdots & \mathbf{x}^{(p)} \end{pmatrix},$$

where $\mathbf{x}^{(j)}$ *is the* j*th column of* $\mathbf{X}$.

Then

$$\mathbf{X}^\top \mathbf{e} = \begin{pmatrix} (\mathbf{x}^{(1)})^\top \mathbf{e} \\ (\mathbf{x}^{(2)})^\top \mathbf{e} \\ \vdots \\ (\mathbf{x}^{(p)})^\top \mathbf{e} \end{pmatrix} = \mathbf{0}.$$

Therefore,

$$(\mathbf{x}^{(j)})^\top \mathbf{e} = 0, \qquad j = 1, \dots, p.$$

Thus, the residual vector is orthogonal to every column of $\mathbf{X}$.

Because the column space $\mathcal{C}(\mathbf{X})$ *consists of all linear combinations of the columns of* $\mathbf{X}$, *it follows that*

$$\boxed{\mathbf{e} \perp \mathcal{C}(\mathbf{X}).}$$

Intuition. The columns of $\mathbf{X}$ represent all directions that the linear model is allowed to use. After least squares has chosen the best fitted value, the remaining error $\mathbf{e}$ has no component left in any of these directions. Otherwise, we could move the fitted value within $\mathcal{C}(\mathbf{X})$ and reduce the residual sum of squares further.

Proposition 2.2 (Fundamental Orthogonal Decomposition)**.** *The fitted response vector is*

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}.$$

Therefore,

$$\boxed{\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X}).}$$

In other words, $\hat{\mathbf{Y}}$ *must lie in the space generated by the columns of* $\mathbf{X}$*.*

Since

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}},$$

we can write

$$\boxed{\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}.}$$

The two components satisfy

$$\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$$

and

$$\mathbf{e} \in \mathcal{C}(\mathbf{X})^{\perp}.$$

Hence,

$$\boxed{\hat{\mathbf{Y}}^{\top}\mathbf{e} = 0.}$$

Geometrically, least squares projects $\mathbf{Y}$ *onto the column space of* $\mathbf{X}$*:*

$$\mathbf{Y} = \underbrace{\hat{\mathbf{Y}}}_{\textit{part explained by the model}} + \underbrace{\mathbf{e}}_{\textit{part orthogonal to the model space}}.$$

Because the two components are orthogonal, the Pythagorean theorem gives

$$\boxed{\|\mathbf{Y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\mathbf{e}\|^2.}$$

This geometric decomposition is the foundation of least squares regression and will later lead naturally to the decomposition of sums of squares in ANOVA.

Exercise 2.1 (Computational Problems)**.** Let

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \mathbf{Y} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

1. Compute $\mathbf{X}^\top\mathbf{X}$.
2. Compute $\mathbf{X}^\top\mathbf{Y}$.
3. Find $\hat{\beta}$.
4. Compute $\hat{\mathbf{Y}}$.
5. Compute the residual vector $\mathbf{e}$.
6. Verify that $\mathbf{X}^\top\mathbf{e} = \mathbf{0}$.

Solution 2.1 (Solution for Exercise 2.1)*.* **Hand calculation**

We have

$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}.$$

Also,

$$\mathbf{X}^\top\mathbf{Y} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

Therefore,

$$\hat{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y} = \begin{pmatrix} 7/6 \\ 1/2 \end{pmatrix}.$$

The fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta} = \begin{pmatrix} 7/6 \\ 5/3 \\ 13/6 \end{pmatrix}.$$

Thus, the residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = \begin{pmatrix} -1/6 \\ 1/3 \\ -1/6 \end{pmatrix}.$$

R calculation

```r
# Define X and Y
X <- matrix(
  c(
    1, 0,
    1, 1,
    1, 2
  ),
  nrow = 3,
  byrow = TRUE
)

Y <- matrix(
  c(1, 2, 2),
  ncol = 1
)

# 1. Compute X'X
XtX <- t(X) %*% X
XtX
```

```
     [,1] [,2]
[1,]    3    3
[2,]    3    5
```

```r
# 2. Compute X'Y
XtY <- t(X) %*% Y
XtY
```

```
     [,1]
[1,]    5
[2,]    6
```

```r
# 3. Compute beta_hat
beta_hat <- solve(XtX) %*% XtY
beta_hat
```

```
         [,1]
[1,] 1.166667
[2,] 0.500000
```

```r
# 4. Compute fitted values
Y_hat <- X %*% beta_hat
Y_hat
```

```
         [,1]
[1,] 1.166667
[2,] 1.666667
[3,] 2.166667
```

```r
# 5. Compute residuals
e <- Y - Y_hat
e
```

```
           [,1]
[1,] -0.1666667
[2,]  0.3333333
[3,] -0.1666667
```

```r
# Optional checks
t(X) %*% e
```

```
              [,1]
[1,] -8.881784e-16
[2,] -8.881784e-16
```

```r
t(Y_hat) %*% e
```

```
              [,1]
[1,] -1.474129e-15
```

Exercise 2.2. Explain in words what the column space

$$\mathcal{C}(\mathbf{X})$$

represents in a linear regression model.

Solution 2.2. The column space of $\mathbf{X}$ is the set of all response patterns that can be represented by the linear model.

More intuitively,

$\mathcal{C}(\mathbf{X})$ describes everything the model is capable of fitting. Anything outside this space cannot be explained by the predictors in $\mathbf{X}$.

2.3.2 Consequence when an Intercept Is Included

If the model includes an intercept, then one column of $\mathbf{X}$ is

$$\mathbf{1}_n = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}.$$

Since the residual vector is orthogonal to every column of $\mathbf{X}$,

$$\mathbf{1}_n^\top \mathbf{e} = 0.$$

Therefore,

$$\boxed{\sum_{i=1}^{n} e_i = 0.}$$

Since

$$e_i = Y_i - \hat{Y}_i,$$

we also have

$$\sum_{i=1}^{n} Y_i = \sum_{i=1}^{n} \hat{Y}_i.$$

Thus,

$$\boxed{\bar{\hat{Y}} = \bar{Y}.}$$

> **Intuition for the Mean of the Fitted Values**
>
> The intercept allows the fitted model to shift up or down. At the least squares solution, there can be no overall tendency for the fitted values to be systematically too high or too low. Therefore, the residuals sum to zero and the fitted values have the same mean as the observed responses.

2.4 The Hat Matrix

In the full-column-rank case,

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}.$$

Substituting

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{Y},$$

we obtain

$$\hat{\mathbf{Y}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{Y}.$$

Define

$$\boxed{\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top.}$$

Then

$$\boxed{\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}.}$$

The matrix $\mathbf{H}$ is called the **hat matrix** because it puts the "hat" on $\mathbf{Y}$:

$$\mathbf{Y} \quad \longrightarrow \quad \mathbf{H}\mathbf{Y} = \hat{\mathbf{Y}}.$$

Proposition 2.3 (Properties of the Hat Matrix)**.** *When* $\mathbf{X}$ *has full column rank, the hat matrix satisfies:*

i) Symmetry

$$\boxed{\mathbf{H}^\top = \mathbf{H}.}$$

ii) Idempotence

$$\boxed{\mathbf{H}^2 = \mathbf{H}.}$$

iii) Projection of the design matrix

$$\boxed{\mathbf{H}\mathbf{X} = \mathbf{X}.}$$

iv) Rank

$$\boxed{\operatorname{rank}(\mathbf{H}) = p.}$$

v) Trace

$$\boxed{\operatorname{tr}(\mathbf{H}) = p.}$$

A symmetric and idempotent matrix represents an orthogonal projection.

Therefore, $\mathbf{H}$ *is the orthogonal projection matrix onto*

$$\mathcal{C}(\mathbf{X}).$$

ℹ Why it is symmetric

We have

$$\begin{aligned}
\mathbf{H}^\top &= \left[\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\right]^\top \\
&= \mathbf{X}\left[(\mathbf{X}^\top\mathbf{X})^{-1}\right]^\top \mathbf{X}^\top.
\end{aligned}$$

Since $\mathbf{X}^\top\mathbf{X}$ is symmetric, its inverse is also symmetric. Therefore,

$$\mathbf{H}^\top = \mathbf{H}.$$

Why it is idempotent

We have

$$\begin{aligned}\mathbf{H}^2 &= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \\ &= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \\ &= \mathbf{H}.\end{aligned}$$

Thus,

$$\mathbf{H}^2 = \mathbf{H}.$$

What Does Idempotence Mean Geometrically?

If we project $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$ once, we obtain

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}.$$

Projecting the fitted vector again does nothing:

$$\mathbf{H}\hat{\mathbf{Y}} = \mathbf{H}^2\mathbf{Y} = \mathbf{H}\mathbf{Y} = \hat{\mathbf{Y}}.$$

Once a vector is already inside the model space, projecting it onto the same space leaves it unchanged.

Notice that, $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$ and $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$, we have

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}.$$

Definition 2.4 (Residual-Maker Matrix)**.** Define

$$\boxed{\mathbf{M} = \mathbf{I}_n - \mathbf{H}.}$$

Then

$$\boxed{\mathbf{e} = \mathbf{M}\mathbf{Y}.}$$

The matrix $\mathbf{M}$ is called the **residual-maker matrix**.

It removes the component of $\mathbf{Y}$ lying in $\mathcal{C}(\mathbf{X})$.

Proposition 2.4 (Properties of the Residual-Maker Matrix)**.** *Since*

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H},$$

we obtain

$$\mathbf{M}^\top = \mathbf{M}$$

and

$$\mathbf{M}^2 = \mathbf{M}.$$

Thus, $\mathbf{M}$ *is also an orthogonal projection matrix.*
While $\mathbf{H}$ *projects onto*

$$\mathcal{C}(\mathbf{X}),$$

$\mathbf{M}$ *projects onto*

$$\mathcal{C}(\mathbf{X})^\perp.$$

In addition,

$$\boxed{\mathbf{HM} = \mathbf{MH} = \mathbf{0}.}$$

Under full column rank,

$$\operatorname{rank}(\mathbf{M}) = n - p$$

and

$$\operatorname{tr}(\mathbf{M}) = n - p.$$

Two Complementary Projection Matrices

The matrices $\mathbf{H}$ and $\mathbf{M}$ split $\mathbb{R}^n$ into two orthogonal components:

$$\mathbb{R}^n = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\mathbf{X})^{\perp}.$$

For the observed response,

$$\mathbf{Y} = \underbrace{\mathbf{HY}}_{\hat{\mathbf{Y}}} + \underbrace{\mathbf{MY}}_{\mathbf{e}}.$$

Thus,

$$\boxed{\mathbf{H} + \mathbf{M} = \mathbf{I}_n.}$$

2.5 The common name to refer to what we have done – Ordinary Least Squares (OLS)

Now that we have seen both the algebraic and geometric views of least squares, we can explain the name **ordinary least squares**, or **OLS**.

- **Squares:** we measure the discrepancy using squared residuals,

$$\sum_{i=1}^{n} e_i^2 = \|\mathbf{Y} - \mathbf{X}\beta\|_2^2.$$

- **Least:** we choose β to make this quantity as small as possible.
- **Ordinary:** we use the usual unweighted sum of squared residuals, so each observation enters the criterion in the same way.

Thus,

$$\boxed{\text{OLS} = \text{minimize the ordinary sum of squared residuals.}}$$

Geometrically, OLS finds the point in $\mathcal{C}(\mathbf{X})$ that is closest to $\mathbf{Y}$ using ordinary Euclidean distance.

Later, we will consider extensions such as weighted least squares and generalized least squares.

Exercise 2.3 (Quick Check)**.** What do **ordinary**, **least**, and **squares** each mean in OLS?

2.6 Sum of Squares Decomposition

2.6.1 Uncentered Geometric Decomposition

Since

$$\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$$

and

$$\hat{\mathbf{Y}}^\top \mathbf{e} = 0,$$

we have

$$\begin{aligned}
\mathbf{Y}^\top \mathbf{Y} &= (\hat{\mathbf{Y}} + \mathbf{e})^\top (\hat{\mathbf{Y}} + \mathbf{e}) \\
&= \hat{\mathbf{Y}}^\top \hat{\mathbf{Y}} + 2\hat{\mathbf{Y}}^\top \mathbf{e} + \mathbf{e}^\top \mathbf{e} \\
&= \hat{\mathbf{Y}}^\top \hat{\mathbf{Y}} + \mathbf{e}^\top \mathbf{e}.
\end{aligned}$$

Therefore,

$$\boxed{\|\mathbf{Y}\|_2^2 = \|\hat{\mathbf{Y}}\|_2^2 + \|\mathbf{e}\|_2^2.}$$

This is a direct application of the Pythagorean theorem.

2.6.2 Centered Sum of Squares Decomposition

When the regression model contains an intercept,

$$\bar{\hat{Y}} = \bar{Y}.$$

Therefore,

$$Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + (Y_i - \hat{Y}_i).$$

The two components are orthogonal, giving

$$\boxed{\sum_{i=1}^{n}(Y_i - \bar{Y})^2 = \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2.}$$

These quantities will later be called

- **total sum of squares**
$$\text{SST} = \sum_{i=1}^{n}(Y_i - \bar{Y})^2;$$
- **regression sum of squares**
$$\text{SSR} = \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2;$$
- **error sum of squares**
$$\text{SSE} = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2.$$

Thus,

$$\boxed{\text{SST} = \text{SSR} + \text{SSE}.}$$

! Centered vs. Uncentered Decomposition

The geometric identity

$$\mathbf{Y}^\top \mathbf{Y} = \hat{\mathbf{Y}}^\top \hat{\mathbf{Y}} + \mathbf{e}^\top \mathbf{e}$$

is an **uncentered** decomposition.
The familiar regression decomposition

$$\text{SST} = \text{SSR} + \text{SSE}$$

is a **centered** decomposition and relies on the model containing an intercept.
These two decompositions should not be confused.

2.6.3 Example: Simple Linear Regression

Consider the simple regression dataset

$$\begin{array}{c|cccc} x_i & 0 & 1 & 2 & 3 \\ \hline y_i & 1 & 3 & 3 & 5 \end{array}.$$

We fit the model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

The response vector is

$$\mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix},$$

and the design matrix is

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}.$$

Step 1: Compute $\mathbf{X}^\top \mathbf{X}$

We have

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}.$$

Step 2: Compute $\mathbf{X}^\top \mathbf{Y}$

We obtain

$$\mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 12 \\ 24 \end{pmatrix}.$$

Step 3: Compute $\hat{\beta}$

Since

$$(\mathbf{X}^\top\mathbf{X})^{-1} = \frac{1}{20}\begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix},$$

the least squares estimator is

$$\begin{aligned}
\hat{\beta} &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y} \\
&= \frac{1}{20}\begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix}\begin{pmatrix} 12 \\ 24 \end{pmatrix} \\
&= \begin{pmatrix} 1.2 \\ 1.2 \end{pmatrix}.
\end{aligned}$$

Thus,

$$\hat{\beta}_0 = 1.2, \qquad \hat{\beta}_1 = 1.2,$$

and the fitted regression line is

$$\boxed{\hat{Y} = 1.2 + 1.2x.}$$

Step 4: Compute the Fitted Values

The fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta} = \begin{pmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{pmatrix}.$$

Step 5: Compute the Residuals

The residual vector is

$$
\begin{aligned}
\mathbf{e} &= \mathbf{Y} - \hat{\mathbf{Y}} \\
&= \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix} - \begin{pmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{pmatrix} \\
&= \begin{pmatrix} -0.2 \\ 0.6 \\ -0.6 \\ 0.2 \end{pmatrix}.
\end{aligned}
$$

Notice that

$$\sum_{i=1}^{4} e_i = -0.2 + 0.6 - 0.6 + 0.2 = 0.$$

This occurs because the model contains an intercept.

Step 6: Verify Orthogonality

We compute

$$\mathbf{X}^\top \mathbf{e} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} -0.2 \\ 0.6 \\ -0.6 \\ 0.2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Thus,

$$\boxed{\mathbf{X}^\top \mathbf{e} = \mathbf{0}.}$$

We can also verify that

$\hat{\mathbf{Y}}^\top \mathbf{e} = 0.$

Step 7: Compute the Hat Matrix

The hat matrix is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top.$$

For this example,

$$
\mathbf{H} = \begin{pmatrix} 0.7 & 0.4 & 0.1 & -0.2 \\ 0.4 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0.2 & 0.3 & 0.4 \\ -0.2 & 0.1 & 0.4 & 0.7 \end{pmatrix}.
$$

Multiplying by $\mathbf{Y}$ gives

$$
\mathbf{H}\mathbf{Y} = \begin{pmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{pmatrix} = \hat{\mathbf{Y}}.
$$

Thus, the hat matrix projects the observed response onto the model space.

We can verify the previous calculations in R.

```r
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

df <- data.frame(
  x = x,
  y = y
)

fit <- lm(y ~ x, data = df)

coef(fit)
```

```
(Intercept)           x
        1.2         1.2
```

The estimated coefficients are

$$
\hat{\beta}_0 = 1.2, \qquad \hat{\beta}_1 = 1.2.
$$

We can also compute the estimator directly using matrix operations.

```r
X <- model.matrix(fit)
X
```

```
  (Intercept) x
1           1 0
2           1 1
3           1 2
4           1 3
attr(,"assign")
[1] 0 1
```

```
Y <- matrix(y, ncol = 1)

beta_hat <-
  solve(t(X) %*% X) %*%
  t(X) %*%
  Y

beta_hat
```

```
            [,1]
(Intercept)  1.2
x            1.2
```

Compute the fitted values and residuals:

```
y_hat <- X %*% beta_hat
e <- Y - y_hat

y_hat
```

```
  [,1]
1  1.2
2  2.4
3  3.6
4  4.8
```

```
e
```

```
  [,1]
1 -0.2
2  0.6
3 -0.6
4  0.2
```

Check that the residuals sum to zero:

```r
sum(e)
```

```
[1] -3.774758e-15
```

Compute the hat matrix:

```r
H <-
  X %*%
  solve(t(X) %*% X) %*%
  t(X)

round(H, 4)
```

```
     1   2   3    4
1  0.7 0.4 0.1 -0.2
2  0.4 0.3 0.2  0.1
3  0.1 0.2 0.3  0.4
4 -0.2 0.1 0.4  0.7
```

Verify symmetry:

```r
round(H - t(H), 10)
```

```
  1 2 3 4
1 0 0 0 0
2 0 0 0 0
3 0 0 0 0
4 0 0 0 0
```

Verify idempotence:

```r
round(H %*% H - H, 10)
```

```
  1 2 3 4
1 0 0 0 0
2 0 0 0 0
3 0 0 0 0
4 0 0 0 0
```

Verify the normal equations geometrically:

```
round(t(X) %*% e, 10)
```

```
            [,1]
(Intercept)    0
x              0
```

Verify that fitted values and residuals are orthogonal:

```
round(t(y_hat) %*% e, 10)
```

```
     [,1]
[1,]    0
```

Construct the residual-maker matrix:

```
M <- diag(nrow(X)) - H

round(M, 4)
```

```
     1    2    3    4
1  0.3 -0.4 -0.1  0.2
2 -0.4  0.7 -0.2 -0.1
3 -0.1 -0.2  0.7 -0.4
4  0.2 -0.1 -0.4  0.3
```

```
round(M %*% M - M, 10)
```

```
  1 2 3 4
1 0 0 0 0
2 0 0 0 0
3 0 0 0 0
4 0 0 0 0
```

Verify that

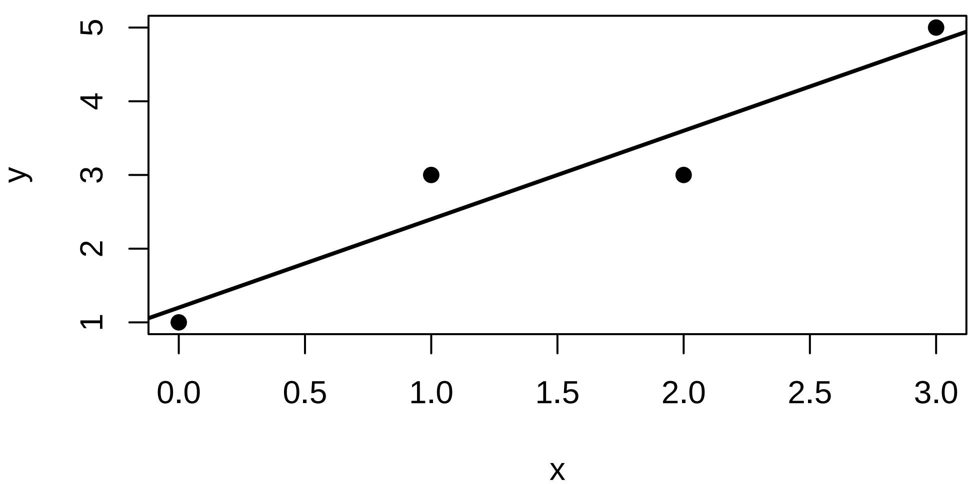
$$\mathbf{e} = \mathbf{M}\mathbf{Y}.$$

```
round(M %*% Y - e, 10)
```

```
  [,1]
1    0
2    0
3    0
4    0
```

Finally, plot the fitted regression line:

```
plot(
  x,
  y,
  pch = 19,
  xlab = "x",
  ylab = "y"
)

abline(
  fit,
  lwd = 2
)
```



2.7 What Happens Without Full Column Rank?

Suppose

$$\operatorname{rank}(\mathbf{X}) < p.$$

Then the columns of $\mathbf{X}$ are linearly dependent, and

$$\mathbf{X}^\top \mathbf{X}$$

is singular.

Therefore,

$$(\mathbf{X}^\top \mathbf{X})^{-1}$$

does not exist.

! A Subtle but Important Point

When $\mathbf{X}$ does not have full column rank,

- a least squares solution still exists;
- $\hat{\beta}$ may not be unique;
- the fitted vector $\hat{\mathbf{Y}}$ is still unique;
- the residual vector $\mathbf{e}$ is still unique.

Thus,

$$\boxed{\hat{\beta} \text{ may not be unique, but } \hat{\mathbf{Y}} \text{ is unique.}}$$

The reason is geometric: even if several coefficient vectors represent the same point in $\mathcal{C}(\mathbf{X})$, the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$ is unique.
We will discuss rank deficiency, generalized inverses, and estimability in more detail later.

2.8 Interpretation of the Geometry

The observed vector

$$\mathbf{Y} \in \mathbb{R}^n.$$

When $\mathbf{X}$ has full column rank p, the model space

$$\mathcal{C}(\mathbf{X})$$

is a p-dimensional subspace of $\mathbb{R}^n$.

Least squares finds the point in this subspace that is closest to $\mathbf{Y}$.

The central relationships are

$$\boxed{\min_{\beta} \|\mathbf{Y} - \mathbf{X}\beta\|_2^2}$$

$$\Downarrow$$

$$\boxed{\mathbf{X}^\top(\mathbf{Y} - \mathbf{X}\widehat{\beta}) = \mathbf{0}}$$

$$\Downarrow$$

$$\boxed{\mathbf{e} \perp \mathcal{C}(\mathbf{X})}$$

$$\Downarrow$$

$$\boxed{\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}}$$

$$\Downarrow$$

$$\boxed{\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}.}$$

Thus,

- fitted values are orthogonal projections;
- residuals are orthogonal to the model space;
- the hat matrix is a projection matrix;
- the residual-maker matrix projects onto the orthogonal complement;
- sum-of-squares decompositions are consequences of the Pythagorean theorem.

2.9 Looking Ahead: Statistical Properties of OLS

Looking Ahead

Everything we have done so far is primarily **algebraic and geometric**.
We defined $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ by solving an optimization problem.
However, under the statistical model, $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$, the response $\mathbf{Y}$ is random. Therefore, $\hat{\beta}$ is also a random vector.
In next chapter, we will study quantities such as

$$\mathbb{E}(\hat{\beta}), \quad \text{and} \quad \mathrm{Var}(\hat{\beta}),$$

followed by estimation of σ^2, sampling distributions, confidence intervals, and hypothesis testing.

Summary

In this chapter, we introduced the least squares estimator through the optimization problem

$$\hat{\beta} = \arg\min_{\beta} \|\mathbf{Y} - \mathbf{X}\beta\|_2^2 .$$

Differentiating the residual sum of squares leads to the normal equations

$$\boxed{\mathbf{X}^\top \mathbf{X}\hat{\beta} = \mathbf{X}^\top \mathbf{Y}.}$$

When $\mathbf{X}$ has full column rank,

$$\boxed{\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.}$$

The fitted response is

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y},$$

where

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$$

is the hat matrix.
Geometrically,

$$\hat{\mathbf{Y}}$$

is the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$.

The residual vector

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$$

lies in the orthogonal complement of the model space, so

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

Therefore,

$$\boxed{\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}}$$

is an orthogonal decomposition.
In next chapter, we will treat $\widehat{\beta}$ as a random vector and study its expectation, variance, sampling distribution, and the foundations of statistical inference for linear models.

2.10 Practice at Home

1. Explain why

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}$$

means that the residual vector is orthogonal to the column space $\mathcal{C}(\mathbf{X})$.

2. Explain why

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$$

is an orthogonal projection, and why

$$\mathbf{H}^2 = \mathbf{H}.$$

3. If the model contains an intercept, explain why

$$\sum_{i=1}^{n} e_i = 0$$

and hence

$$\bar{\hat{Y}} = \bar{Y}.$$

4. Let

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \mathbf{Y} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

   Compute

$$\hat{\beta}, \qquad \hat{\mathbf{Y}}, \qquad \mathbf{e},$$

   and verify that

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

5. Show that the hat matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top$$

   is symmetric and idempotent.

Side Fact: Matrix Calculus Facts Used in this chapter

Let β be a column vector.

For a constant vector $\mathbf{a}$,

$$\frac{\partial}{\partial \beta}\left(\mathbf{a}^\top \beta\right) = \mathbf{a}.$$

For a constant matrix $\mathbf{A}$,

$$\frac{\partial}{\partial \beta}\left(\beta^\top \mathbf{A} \beta\right) = (\mathbf{A} + \mathbf{A}^\top)\beta.$$

If $\mathbf{A}$ is symmetric,

$$\mathbf{A}^\top = \mathbf{A},$$

and therefore

$$\boxed{\frac{\partial}{\partial \beta}\left(\beta^\top \mathbf{A} \beta\right) = 2\mathbf{A}\beta.}$$

These identities justify the derivative used in the least squares criterion.

3 Distribution Theory of OLS and Inference

In the previous chapter, we studied least squares primarily from an **algebraic and geometric** point of view.

We obtained

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y},$$

and showed that

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}, \qquad \mathbf{e} = \mathbf{M}\mathbf{Y},$$

where

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$$

and

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

We now move from **estimation** to **statistical inference**.

Since $\mathbf{Y}$ is random, quantities such as

$$\hat{\beta}, \qquad \hat{\mathbf{Y}}, \qquad \mathbf{e}, \qquad \text{SSE}$$

are also random.

Our goal is to understand their sampling distributions and use them to quantify uncertainty.

> Learning Objectives
>
> By the end of this chapter, students should be able to:
>
> - derive the expectation and variance of the OLS estimator;
> - state the normal linear model;

- derive the distribution of the OLS estimator under normality;
- explain the distribution of the residual sum of squares;
- explain why the residual degrees of freedom are $n - p$;
- estimate σ^2 using the mean squared error;
- explain why $\hat{\beta}$ and SSE are independent;
- construct confidence intervals for regression coefficients;
- perform hypothesis tests for regression coefficients;
- perform inference for linear combinations of regression coefficients;
- distinguish inference for a mean response from prediction of a future observation;
- interpret standard regression output from R.

Reading

Recommended reading for this chapter:

- Seber and Lee:
  - sections on distribution theory of least squares estimators;
  - estimation of the error variance;
  - inference for regression coefficients.
- Montgomery, Peck, and Vining:
  - sections on confidence intervals;
  - hypothesis testing;
  - estimation and prediction in linear regression.

Recap from Least Squares

From Chapter 2, when $\mathbf{X}$ has full column rank,

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

We also have

$$\hat{\mathbf{Y}} = \mathbf{HY}, \qquad \mathbf{e} = \mathbf{MY},$$

with

$$\mathbf{H} + \mathbf{M} = \mathbf{I}_n$$

and

$$\mathbf{HM} = \mathbf{MH} = \mathbf{0}.$$

Geometrically,

$$
\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}
$$

is an orthogonal decomposition.
In this chapter, we will see that this geometric decomposition also provides the foundation for regression inference.

❗ Throughout This Chapter

We treat the design matrix $\mathbf{X}$ as fixed.
If the predictors are random, the results can instead be interpreted conditionally on the observed design matrix $\mathbf{X}$.

3.1 Mean and Variance of the OLS Estimator

Before introducing normality, we can already study the expectation and variance of the OLS estimator.

Assume

$$
\mathbb{E}(\varepsilon) = \mathbf{0}
$$

and

$$
\mathrm{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n .
$$

Recall that

$$
\mathbf{Y} = \mathbf{X}\beta + \varepsilon .
$$

Substituting this into the OLS estimator gives

$$
\begin{aligned}
\hat{\beta} &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} \\
&= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\beta + \varepsilon) \\
&= \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon .
\end{aligned}
$$

Thus,

$$\boxed{\hat{\beta} = \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon.}$$

This representation will be useful throughout the chapter.

Proposition 3.1 (Mean and Variance of OLS)**.** *Under*

$$\mathbb{E}(\varepsilon) = \mathbf{0}$$

and

$$\mathrm{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n,$$

the OLS estimator satisfies

$$\boxed{\mathbb{E}\left(\hat{\beta}\right) = \beta}$$

and

$$\boxed{\mathrm{Var}\left(\hat{\beta}\right) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.}$$

3.1.1 Why Is OLS Unbiased?

Using

$$\hat{\beta} = \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon,$$

we obtain

$$\begin{aligned} \mathbb{E}\left(\hat{\beta}\right) &= \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbb{E}\left(\varepsilon\right) \\ &= \beta. \end{aligned}$$

Thus,

$$\boxed{\mathbb{E}\left(\hat{\beta}\right) = \beta.}$$

OLS is therefore an **unbiased estimator** of β.

Intuitively, if we repeated the experiment many times using the same design matrix, the OLS estimates would fluctuate around the true parameter vector.

3.1.2 Variance of OLS

Similarly,

$$\begin{aligned}
\operatorname{Var}\left(\hat{\beta}\right) &= \operatorname{Var}\left[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\varepsilon\right] \\
&= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \operatorname{Var}(\varepsilon)\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} \\
&= \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.
\end{aligned}$$

> **!** Normality Is Not Needed Yet
>
> The results
>
> $$\mathbb{E}\left(\hat{\beta}\right) = \beta$$
>
> and
>
> $$\operatorname{Var}\left(\hat{\beta}\right) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$$
>
> do **not** require normally distributed errors.
> They follow from the first two moment assumptions
>
> $$\mathbb{E}(\varepsilon) = \mathbf{0}$$
>
> and
>
> $$\operatorname{Var}(\varepsilon) = \sigma^2\mathbf{I}_n.$$
>
> Normality becomes important when we want exact finite sample distributions and inference.

Exercise 3.1 (Which Assumptions Are Needed?)**.** Suppose

$$\mathbb{E}(\varepsilon) = \mathbf{0}$$

and

$$\mathrm{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n.$$

Do we need to assume that the errors are normally distributed to conclude that

$$\mathbb{E}\left(\widehat{\beta}\right) = \beta?$$

Solution 3.1. No.

Normality is **not** required for unbiasedness.

The result follows from

$$\widehat{\beta} = \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon$$

and

$$\mathbb{E}(\varepsilon) = \mathbf{0}.$$

Normality will be needed later for exact finite sample inference.

3.2 The Normal Linear Model

To obtain exact sampling distributions, we now strengthen the assumptions.

Definition 3.1 (Normal Linear Model)**.** The **normal linear model** assumes

$$\boxed{\mathbf{Y} \sim N_n\left(\mathbf{X}\beta, \sigma^2 \mathbf{I}_n\right).}$$

Equivalently,

$$\boxed{\varepsilon \sim N_n\left(\mathbf{0}, \sigma^2 \mathbf{I}_n\right).}$$

Under this assumption:

- each response has a normal distribution;
- the errors have common variance σ^2;
- the errors are independent;
- exact finite sample distributions can be obtained for many regression statistics.

Why Does Normality Matter?

Least squares itself does not require normality.
Normality is introduced because it gives us exact distributions for

$$\hat{\beta}, \qquad \text{SSE}, \qquad T, \qquad F.$$

These distributions allow us to construct exact finite sample confidence intervals and hypothesis tests.

Exercise 3.2 (What Changes When We Add Normality?)**.** Suppose the errors satisfy

$$\mathbb{E}(\varepsilon) = \mathbf{0}, \qquad \operatorname{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n,$$

but are not normally distributed.

Which of the following can still hold?

1. OLS can still be computed.
2. OLS can still be unbiased.
3. We can still have

$$\operatorname{Var}\left(\hat{\beta}\right) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

4. Exact finite sample t and F distributions automatically hold.

Solution 3.2. Statements 1, 2, and 3 can still hold.

Statement 4 does not generally hold.

Normality is not required to define OLS or derive its mean and variance. Its main role here is to provide exact finite sample distributions for inference.

3.3 Distribution of the OLS Estimator

Recall that

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

Thus, $\hat{\beta}$ is a **linear transformation** of $\mathbf{Y}$.

A fundamental property of the multivariate normal distribution is that a linear transformation of a multivariate normal vector is again multivariate normal.

Theorem 3.1 (Distribution of the OLS Estimator)**.** *Under the normal linear model,*

$$\boxed{\hat{\beta} \sim N_p\left(\beta, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}\right).}$$

Let

$$\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}.$$

Then

$$\operatorname{Var}\left(\hat{\beta}\right) = \sigma^2 \mathbf{C}.$$

If c_{jj} denotes the jth diagonal element of $\mathbf{C}$, then

$$\hat{\beta}_j \sim N\left(\beta_j, \sigma^2 c_{jj}\right).$$

Therefore,

$$\boxed{\frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{c_{jj}}} \sim N(0,1).}$$

Exercise 3.3 (Reading the Covariance Matrix)**.** Let

$$\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}.$$

If c_{jj} is the jth diagonal element of $\mathbf{C}$:

1. What is $\operatorname{Var}(\hat{\beta}_j)$?
2. What happens to the uncertainty of $\hat{\beta}_j$ if c_{jj} becomes larger?

Solution 3.3. The variance is

$$\boxed{\operatorname{Var}(\hat{\beta}_j) = \sigma^2 c_{jj}.}$$

Therefore, a larger c_{jj} means greater sampling variability and less precise estimation of β_j.

Thus, the precision of OLS depends not only on σ^2, but also on the design matrix $\mathbf{X}$.

There is still one difficulty.

The quantity

$$\frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{c_{jj}}}$$

depends on the unknown parameter σ.

Before performing inference, we need to estimate σ^2.

Exercise 3.4 (Why Can We Not Use the Normal Statistic Directly?)**.** We have

$$\frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{c_{jj}}} \sim N(0, 1).$$

Why can we usually not use this statistic directly for inference?

Solution 3.4. Because the error standard deviation σ is unknown.

We therefore need to estimate σ from the data.

Replacing σ with an estimator will lead naturally to the t distribution.

3.4 Residual Sum of Squares and Estimation of σ^2

Recall from Chapter 2 that

$$\mathbf{e} = \mathbf{M}\mathbf{Y},$$

where

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

Since

$$\mathbf{M}\mathbf{X} = \mathbf{0},$$

we have

$$\begin{aligned}\mathbf{e} &= \mathbf{M}\mathbf{Y}\\ &= \mathbf{M}\left(\mathbf{X}\beta + \varepsilon\right)\\ &= \mathbf{M}\varepsilon.\end{aligned}$$

Thus,

$$\boxed{\mathbf{e} = \mathbf{M}\varepsilon.}$$

The residual vector therefore contains the part of the random error that lies outside the model space.

3.4.1 Residual Sum of Squares

The residual sum of squares is

$$\mathrm{SSE} = \mathbf{e}^\top \mathbf{e}.$$

Since

$$\mathbf{e} = \mathbf{M}\varepsilon,$$

and $\mathbf{M}$ is symmetric and idempotent,

$$\begin{aligned}\mathrm{SSE} &= \varepsilon^\top \mathbf{M}^\top \mathbf{M}\varepsilon \\ &= \varepsilon^\top \mathbf{M}\varepsilon.\end{aligned}$$

Therefore,

$$\boxed{\mathrm{SSE} = \varepsilon^\top \mathbf{M}\varepsilon.}$$

3.4.2 Distribution of SSE

Under the normal linear model,

$$\frac{\varepsilon}{\sigma} \sim N_n\left(\mathbf{0}, \mathbf{I}_n\right).$$

Let

$$\mathbf{Z} = \frac{\varepsilon}{\sigma}.$$

Then

$$\frac{\mathrm{SSE}}{\sigma^2} = \mathbf{Z}^\top \mathbf{M}\mathbf{Z}.$$

Recall that $\mathbf{M}$ is symmetric and idempotent and

$$\operatorname{rank}(\mathbf{M}) = n - p.$$

Therefore,

Theorem 3.2 (Distribution of the Residual Sum of Squares)**.** *Under the normal linear model,*

$$\boxed{\frac{\mathrm{SSE}}{\sigma^2} \sim \chi^2_{n-p}.}$$

3.4.3 Why Are the Degrees of Freedom $n - p$?

Recall that

$$\text{rank}(\mathbf{H}) = p$$

and

$$\text{rank}(\mathbf{M}) = n - p.$$

The fitted vector lies in

$$\mathcal{C}(\mathbf{X}),$$

which has dimension p.

The residual vector lies in

$$\mathcal{C}(\mathbf{X})^{\perp},$$

which has dimension

$$n - p.$$

Therefore, the residual variation has

$$\boxed{n - p}$$

degrees of freedom.

Geometric Interpretation of Degrees of Freedom

The response space can be decomposed as

$$\mathbb{R}^n = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\mathbf{X})^{\perp}.$$

The fitted component occupies p dimensions.
The remaining

$$n - p$$

dimensions are available for residual variation.

> This provides a geometric explanation for why estimating p regression parameters costs p degrees of freedom.

Exercise 3.5 (Degrees of Freedom)**.** Suppose a regression model has

$$n = 20$$

observations and

$$p = 3$$

regression parameters.

1. What is the residual degrees of freedom?
2. Why is it not 20?

Solution 3.5. The residual degrees of freedom are

$$n - p = 20 - 3 = 17.$$

It is not 20 because fitting the regression model uses $p = 3$ dimensions of the response space. The residual vector therefore lies in a space of dimension

$$20 - 3 = 17.$$

3.4.4 Estimating σ^2

Since a chi-square random variable with ν degrees of freedom has expectation ν,

$$\mathbb{E}\left(\frac{\text{SSE}}{\sigma^2}\right) = n - p.$$

Therefore,

$$\mathbb{E}(\text{SSE}) = (n - p)\sigma^2.$$

This gives us an unbiased estimator of the error variance.

Definition 3.2 (Mean Squared Error)**.** The **mean squared error** is defined as

$$\boxed{\mathrm{MSE} = \hat{\sigma}^2 = \frac{\mathrm{SSE}}{n-p}.}$$

It satisfies

$$\mathbb{E}\left(\hat{\sigma}^2\right) = \sigma^2.$$

The estimated error standard deviation is

$$\boxed{\hat{\sigma} = \sqrt{\mathrm{MSE}}.}$$

Exercise 3.6 (Estimate the Error Variance)**.** Suppose

$$n = 20, \qquad p = 3, \qquad \mathrm{SSE} = 34.$$

Compute

$$\hat{\sigma}^2$$

and

$$\hat{\sigma}.$$

Solution 3.6. The residual degrees of freedom are

$$n - p = 17.$$

Therefore,

$$\hat{\sigma}^2 = \frac{34}{17} = 2.$$

Hence,

$$\boxed{\hat{\sigma} = \sqrt{2} \approx 1.414.}$$

3.5 Independence of OLS and Residual Variation

A crucial result under the normal linear model is

$$\boxed{\hat{\beta} \quad \text{and} \quad \text{SSE} \quad \text{are independent.}}$$

This result is closely connected to the projection geometry from the previous chapter.

3.5.1 Why Does Independence Occur?

Recall that

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$$

and

$$\mathbf{e} = \mathbf{M}\mathbf{Y}.$$

We also know that

$$\mathbf{H}\mathbf{M} = \mathbf{0}.$$

Thus, the fitted component and the residual component come from orthogonal parts of the response vector.

> **Geometry Becomes Probability**
>
> In the previous chapter, we learned that
>
> $$\hat{\mathbf{Y}} \perp \mathbf{e}.$$
>
> This is a geometric statement.
> Under the normal linear model, these components are jointly normal.
> For jointly normal random vectors, zero covariance implies independence.
> Thus, under normality, geometric orthogonality leads to probabilistic independence.

More formally,

$$\hat{\beta} - \beta = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \varepsilon$$

and

$$\mathbf{e} = \mathbf{M}\varepsilon.$$

Therefore,

$$\begin{aligned}\operatorname{Cov}\left(\hat{\beta}, \mathbf{e}\right) &= \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{M} \\ &= \mathbf{0},\end{aligned}$$

because

$$\mathbf{X}^\top\mathbf{M} = \mathbf{0}.$$

Since the two random vectors are jointly normal,

$$\hat{\beta} \perp\!\!\!\perp \mathbf{e}.$$

Since

$$\text{SSE} = \mathbf{e}^\top\mathbf{e}$$

is a function of $\mathbf{e}$,

$$\boxed{\hat{\beta} \perp\!\!\!\perp \text{SSE}.}$$

Exercise 3.7 (Does Orthogonal Always Mean Independent?)**.** We know that

$$\mathbf{HM} = \mathbf{0}.$$

Does the orthogonality between the fitted and residual components automatically imply that they are independent?

Why is normality important?

Solution 3.7. No.

In general, zero covariance or orthogonality does **not** imply independence.

Under the normal linear model, the fitted and residual components are jointly normal.

For jointly normal random vectors, zero covariance implies independence.

Thus, normality is what allows us to move from orthogonality to independence.

3.6 Inference for a Single Regression Coefficient

For the jth regression coefficient,

$$\hat{\beta}_j \sim N\left(\beta_j, \sigma^2 c_{jj}\right).$$

If σ were known, we could use

$$\frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{c_{jj}}} \sim N(0,1).$$

In practice, σ is unknown.

We replace it with

$$\hat{\sigma} = \sqrt{\frac{\text{SSE}}{n-p}}.$$

3.6.1 Standard Error

The estimated standard deviation of $\hat{\beta}_j$ is

Definition 3.3 (Standard Error of an OLS Coefficient)**.** The standard error of $\hat{\beta}_j$ is

$$\boxed{\text{SE}\left(\hat{\beta}_j\right) = \hat{\sigma}\sqrt{c_{jj}}.}$$

3.6.2 Why Does the t Distribution Appear?

We know that

$$Z = \frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{c_{jj}}} \sim N(0,1).$$

We also know that

$$U = \frac{\text{SSE}}{\sigma^2} \sim \chi^2_{n-p}.$$

In addition,

$$Z \quad \text{and} \quad U$$

are independent.

Therefore,

$$\frac{Z}{\sqrt{U/(n-p)}} \sim t_{n-p}.$$

Substituting the regression quantities gives

Theorem 3.3 (t Distribution for an OLS Coefficient)**.** *Under the normal linear model,*

$$\boxed{T = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{c_{jj}}} \sim t_{n-p}.}$$

> **Normal vs. t Distribution**
>
> If σ were known, we would use a normal distribution.
> Because σ is unknown and must be estimated from the data, we introduce additional uncertainty.
> The t distribution accounts for this additional uncertainty.
> As the degrees of freedom increase,
>
> $$t_\nu$$
>
> approaches

$$N(0,1).$$

Exercise 3.8 (Why Do We Use a t Distribution?)**.** Suppose

$$\hat{\beta}_j \sim N\left(\beta_j, \sigma^2 c_{jj}\right).$$

Why do we use a t distribution rather than a standard normal distribution when performing inference for β_j?

Solution 3.8. Because σ is unknown.

We replace σ with the estimate $\hat{\sigma}$ obtained from the residual sum of squares.

This additional estimation uncertainty changes the standardized statistic from a normal distribution to a t distribution.

3.6.3 Confidence Interval for β_j

A $100(1-\alpha)\%$ confidence interval for β_j is

$$\boxed{\hat{\beta}_j \pm t_{1-\alpha/2,\, n-p}\, \mathrm{SE}\left(\hat{\beta}_j\right).}$$

Equivalently,

$$\boxed{\hat{\beta}_j \pm t_{1-\alpha/2,\, n-p}\hat{\sigma}\sqrt{c_{jj}}.}$$

3.6.4 Hypothesis Test for β_j

Suppose we want to test

$$H_0 : \beta_j = \beta_{j,0}$$

against

$$H_1 : \beta_j \neq \beta_{j,0}.$$

The test statistic is

$$\boxed{T = \frac{\hat{\beta}_j - \beta_{j,0}}{\mathrm{SE}(\hat{\beta}_j)}.}$$

Under H_0,

$$T \sim t_{n-p}.$$

For a two sided test at significance level α, reject H_0 if

$$|T| > t_{1-\alpha/2,\, n-p}.$$

Exercise 3.9 (Confidence Interval and Hypothesis Test)**.** Suppose we test

$$H_0 : \beta_j = 0$$

against

$$H_1 : \beta_j \neq 0$$

at significance level

$$\alpha = 0.05.$$

What should happen if the corresponding 95% confidence interval for β_j does not contain zero?

Solution 3.9. We reject

$$H_0 : \beta_j = 0$$

at the 5% significance level.

A two sided hypothesis test at level α and a $100(1-\alpha)\%$ confidence interval give equivalent conclusions.

3.7 Inference for Linear Combinations

Sometimes the quantity of interest is not a single coefficient.

Suppose we are interested in

$$\mathbf{a}^\top \beta,$$

where $\mathbf{a}$ is a fixed $p \times 1$ vector.

Since

$$\hat{\beta} \sim N_p\left(\beta, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}\right),$$

we have

$$\mathbf{a}^\top \hat{\beta} \sim N\left(\mathbf{a}^\top \beta, \sigma^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}\right).$$

Therefore,

$$\boxed{\frac{\mathbf{a}^\top \hat{\beta} - \mathbf{a}^\top \beta}{\hat{\sigma}\sqrt{\mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}}} \sim t_{n-p}.}$$

A $100(1-\alpha)\%$ confidence interval for $\mathbf{a}^\top \beta$ is

$$\boxed{\mathbf{a}^\top \hat{\beta} \pm t_{1-\alpha/2,\, n-p}\, \hat{\sigma} \sqrt{\mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}}.}$$

Why Is This Useful?

Many regression questions can be expressed as linear combinations of the regression coefficients.
Examples include:

- a single coefficient;
- the difference between two coefficients;
- the mean response at a particular predictor value.

Thus, inference for a single regression coefficient is a special case of inference for a linear combination.

Exercise 3.10 (A Single Coefficient as a Linear Combination)**.** Suppose

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}.$$

What vector $\mathbf{a}$ would you choose so that

$$\mathbf{a}^\top \beta = \beta_1?$$

Solution 3.10*.* Choose

$$\boxed{\mathbf{a} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.}$$

Then

$$\mathbf{a}^\top \beta = \beta_1.$$

Thus, inference for an individual coefficient is a special case of inference for a linear combination.

3.8 General Linear Hypotheses

Sometimes we want to test several linear restrictions simultaneously.

Suppose

$$H_0 : \mathbf{C}\beta = \mathbf{d},$$

where $\mathbf{C}$ is an $r \times p$ matrix of rank r and $\mathbf{d}$ is an $r \times 1$ vector.

Definition 3.4 (General Linear Hypothesis)**.** A hypothesis of the form

$$H_0 : \mathbf{C}\beta = \mathbf{d}$$

is called a **general linear hypothesis**.

It allows several linear restrictions on the regression coefficients to be tested simultaneously.

The corresponding test statistic is

$$F = \frac{(\mathbf{C}\hat{\beta} - \mathbf{d})^\top \left[\mathbf{C}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{C}^\top\right]^{-1} (\mathbf{C}\hat{\beta} - \mathbf{d})}{r\hat{\sigma}^2}.$$

Under H_0,

$$\boxed{F \sim F_{r,\, n-p}.}$$

3.8.1 Relationship Between t and F

When only one restriction is being tested,

$$r = 1.$$

In this case,

$$\boxed{F = T^2.}$$

Thus, the familiar t test for one coefficient is a special case of the general F test.

We will study F tests more thoroughly in the next chapter when we develop the ANOVA decomposition and compare nested regression models.

Exercise 3.11 (t Test or F Test?)**.** Suppose we want to test only

$$H_0 : \beta_2 = 0.$$

Could we use either a t test or an F test?

How are the two test statistics related?

Solution 3.11*.* Yes.

Because there is only one linear restriction, either test can be used.

The corresponding statistics satisfy

$$\boxed{F = T^2.}$$

Therefore, the two tests produce the same conclusion.

3.9 Inference for the Mean Response

Suppose we want to estimate the mean response at a predictor vector

$$\mathbf{x}_0.$$

The true mean response is

$$\mu(\mathbf{x}_0) = \mathbf{x}_0^\top \beta.$$

Its estimator is

$$\widehat{\mu}(\mathbf{x}_0) = \mathbf{x}_0^\top \widehat{\beta}.$$

Since this is a linear combination of $\widehat{\beta}$,

$$\operatorname{Var}\left(\widehat{\mu}(\mathbf{x}_0)\right) = \sigma^2 \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0.$$

Thus,

$$\frac{\mathbf{x}_0^\top \widehat{\beta} - \mathbf{x}_0^\top \beta}{\hat{\sigma}\sqrt{\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}} \sim t_{n-p}.$$

Therefore, a $100(1-\alpha)\%$ confidence interval for the mean response is

$$\boxed{\mathbf{x}_0^\top \widehat{\beta} \pm t_{1-\alpha/2,\, n-p}\hat{\sigma}\sqrt{\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}.}$$

3.10 Prediction for a New Observation

Now suppose we want to predict one new observation at the same predictor value $\mathbf{x}_0$:

$$Y_{\text{new}} = \mathbf{x}_0^\top \beta + \varepsilon_{\text{new}},$$

where

$$\varepsilon_{\text{new}} \sim N(0, \sigma^2)$$

and is independent of the original data.

The predictor is

$$\widehat{Y}_{\text{new}} = \mathbf{x}_0^\top \widehat{\beta}.$$

The prediction error is

$$Y_{\text{new}} - \mathbf{x}_0^\top \widehat{\beta}.$$

Its variance is

$$\operatorname{Var}\left(Y_{\text{new}} - \mathbf{x}_0^\top \widehat{\beta}\right) = \sigma^2 \left[1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0\right].$$

Therefore, a $100(1-\alpha)\%$ prediction interval is

$$\boxed{\mathbf{x}_0^\top \widehat{\beta} \pm t_{1-\alpha/2,\, n-p}\, \widehat{\sigma} \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}.}$$

! Mean Response vs. New Observation

The two intervals answer different questions.

Goal	Source of uncertainty
Estimate a mean response	Uncertainty in estimating the regression mean
Predict a new observation	Uncertainty in the regression mean plus new random variation

The additional 1 in

$$1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$$

represents the variability of the new observation itself.
Therefore, at the same $\mathbf{x}_0$ and confidence level, the prediction interval is wider than the confidence interval for the mean response.

Exercise 3.12 (Mean Response or Prediction?)**.** Consider the following questions.

1. What is the **average stopping distance** of all cars travelling at 50 mph?

2. What stopping distance should we expect for **one particular future car** travelling at 50 mph?

Which question requires a confidence interval for the mean response, and which requires a prediction interval?

Solution 3.12. The first question concerns the **mean response**, so we use a confidence interval for the regression mean.

The second question concerns **one future observation**, so we use a prediction interval.

The prediction interval is wider because it also includes the random variation associated with the future observation.

3.11 Worked Example

Consider again the simple regression dataset from Chapter 2:

$$\begin{array}{c|cccc} x_i & 0 & 1 & 2 & 3 \\ \hline Y_i & 1 & 3 & 3 & 5 \end{array}.$$

The design matrix and response vector are

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \qquad \mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix}.$$

From the previous chapter,

$$\hat{\beta} = \begin{pmatrix} 1.2 \\ 1.2 \end{pmatrix},$$

and

$$\hat{\mathbf{Y}} = \begin{pmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{pmatrix}.$$

Therefore,

$$\mathbf{e} = \begin{pmatrix} -0.2 \\ 0.6 \\ -0.6 \\ 0.2 \end{pmatrix}.$$

3.11.1 Estimate the Error Variance

The residual sum of squares is

$$\begin{aligned} \text{SSE} &= \mathbf{e}^\top \mathbf{e} \\ &= (-0.2)^2 + 0.6^2 + (-0.6)^2 + 0.2^2 \\ &= 0.8. \end{aligned}$$

Since

$$n = 4$$

and

$$p = 2,$$

the residual degrees of freedom are

$$n - p = 2.$$

Thus,

$$\boxed{\hat{\sigma}^2 = \frac{0.8}{2} = 0.4.}$$

Therefore,

$$\boxed{\hat{\sigma} = \sqrt{0.4} \approx 0.632.}$$

3.11.2 Standard Error of the Slope

We have

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{20} \begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix}.$$

For the slope,

$$c_{22} = \frac{4}{20} = 0.2.$$

Therefore,

$$\mathrm{SE}(\hat{\beta}_1) = \sqrt{0.4(0.2)} = \sqrt{0.08} \approx 0.283.$$

3.11.3 Test the Slope

Consider

$$H_0 : \beta_1 = 0$$

versus

$$H_1 : \beta_1 \neq 0.$$

The test statistic is

$$T = \frac{1.2}{0.283} \approx 4.243.$$

The residual degrees of freedom are

$$2.$$

For a 5% two sided test,

$$t_{0.975,2} \approx 4.303.$$

Since

$$|4.243| < 4.303,$$

we fail to reject H_0 at the 5% significance level.

> **i What Does This Example Show?**
>
> The estimated slope
>
> $$\hat{\beta}_1 = 1.2$$
>
> is relatively large.
> However, there are only four observations, leaving only two residual degrees of freedom.
> Therefore, there is substantial uncertainty in the estimate.
> A large estimated effect does not automatically imply strong evidence against the null hypothesis.

3.11.4 Confidence Interval for the Slope

A 95% confidence interval is

$$1.2 \pm 4.303(0.283),$$

which gives approximately

$$\boxed{(-0.017,\ 2.417).}$$

The interval contains zero, which agrees with the hypothesis test.

Exercise 3.13 (Check the Connection)**.** Without performing another hypothesis test, what conclusion can you draw from the fact that the 95% confidence interval

$$(-0.017,\ 2.417)$$

contains zero?

Solution 3.13. We fail to reject

$$H_0 : \beta_1 = 0$$

at the 5% significance level.

This agrees with the previous t test.

3.11.5 Mean Response at $x_0 = 2$

For simple linear regression with an intercept,

$$\mathbf{x}_0 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

The estimated mean response is

$$\hat{\mu}(\mathbf{x}_0) = \mathbf{x}_0^\top \hat{\beta} = 3.6.$$

Also,

$$\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0 = 0.3.$$

Therefore,

$$\mathrm{SE}\left(\hat{\mu}(\mathbf{x}_0)\right) = \sqrt{0.4(0.3)} \approx 0.346.$$

The 95% confidence interval for the mean response is approximately

$$\boxed{(2.110,\ 5.090).}$$

3.11.6 Prediction at $x_0 = 2$

For one new observation,

$$\mathrm{SE}_{\text{pred}} = \sqrt{0.4(1 + 0.3)} \approx 0.721.$$

Therefore, the 95% prediction interval is approximately

$$\boxed{(0.497,\ 6.703).}$$

Notice that the prediction interval is substantially wider than the confidence interval for the mean response.

3.12 R Demonstration

We can reproduce the previous calculations using R.

```r
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

fit <- lm(y ~ x)

summary(fit)
```

```
Call:
lm(formula = y ~ x)

Residuals:
   1    2    3    4 
-0.2  0.6 -0.6  0.2 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)  
(Intercept)   1.2000     0.5292   2.268   0.1515  
x             1.2000     0.2828   4.243   0.0513 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6325 on 2 degrees of freedom
Multiple R-squared:    0.9, Adjusted R-squared:   0.85 
F-statistic:    18 on 1 and 2 DF,  p-value: 0.05132
```

3.12.1 Estimated Coefficients

```r
coef(fit)
```

```
(Intercept)           x 
        1.2         1.2 
```

3.12.2 Residual Degrees of Freedom and Error Variance

```r
df.residual(fit)
```

```
[1] 2
```

```r
sigma(fit)
```

```
[1] 0.6324555
```

```r
sigma(fit)^2
```

```
[1] 0.4
```

3.12.3 Covariance Matrix and Standard Errors

```r
vcov(fit)
```

```
            (Intercept)     x
(Intercept)        0.28 -0.12
x                 -0.12  0.08
```

```r
sqrt(diag(vcov(fit)))
```

```
(Intercept)           x
  0.5291503   0.2828427
```

3.12.4 Verify the Formulas Using Matrix Operations

```r
X <- model.matrix(fit)
Y <- matrix(y, ncol = 1)

n <- nrow(X)
p <- ncol(X)
```

```r
beta_hat <-
  solve(t(X) %*% X) %*%
  t(X) %*%
  Y

e <- Y - X %*% beta_hat

SSE <- as.numeric(t(e) %*% e)

sigma2_hat <- SSE / (n - p)

V_beta_hat <-
  sigma2_hat *
  solve(t(X) %*% X)

SE_beta_hat <-
  sqrt(diag(V_beta_hat))

beta_hat
```

```
            [,1]
(Intercept)  1.2
x            1.2
```

```r
SSE
```

```
[1] 0.8
```

```r
sigma2_hat
```

```
[1] 0.4
```

```r
V_beta_hat
```

```
            (Intercept)     x
(Intercept)        0.28 -0.12
x                 -0.12  0.08
```

```r
SE_beta_hat
```

```
(Intercept)           x 
  0.5291503   0.2828427 
```

3.12.5 Hypothesis Tests and Confidence Intervals

R automatically reports coefficient level t tests:

```r
summary(fit)$coefficients
```

```
            Estimate Std. Error  t value  Pr(>|t|)
(Intercept)      1.2  0.5291503 2.267787 0.1514719
x                1.2  0.2828427 4.242641 0.0513167
```

Confidence intervals for the regression coefficients can be obtained using

```r
confint(fit)
```

```
                  2.5 %   97.5 %
(Intercept) -1.07674982 3.476750
x           -0.01697397 2.416974
```

3.12.6 Mean Response and Prediction

At

$$x_0 = 2,$$

use

```r
newdat <- data.frame(x = 2)

predict(
  fit,
  newdata = newdat,
  interval = "confidence"
)
```

```
  fit      lwr      upr
1 3.6 2.109517 5.090483
```

```r
predict(
  fit,
  newdata = newdat,
  interval = "prediction"
)
```

```
  fit      lwr      upr
1 3.6 0.497313 6.702687
```

The first interval estimates the **mean response**.

The second interval predicts **one future observation**.

3.13 Reading Standard Regression Output

For

```r
summary(fit)
```

the coefficient table contains four important columns:

Column	Interpretation
`Estimate`	Estimated regression coefficient
`Std. Error`	Estimated standard deviation of the coefficient estimator
`t value`	t statistic for testing whether the coefficient equals zero
`Pr(>|t|)`	Two sided p value for the test

The output also includes:

- the residual standard error;
- the residual degrees of freedom;
- R^2;
- adjusted R^2;
- an overall F test.

We will study R^2, the ANOVA decomposition, and the overall F test more carefully in the next chapter.

Exercise 3.14 (Reading an R Regression Table)**.** Suppose R reports

	Estimate	Std. Error	t value	Pr(>
x_1	2.40	0.60	4.00	0.001

Answer the following.

1. What is $\hat{\beta}_1$?
2. What is $\mathrm{SE}(\hat{\beta}_1)$?
3. What null hypothesis is being tested by default?

Solution 3.14. The estimate is

$$\hat{\beta}_1 = 2.40.$$

The standard error is

$$\mathrm{SE}(\hat{\beta}_1) = 0.60.$$

The default coefficient level test is

$$H_0 : \beta_1 = 0.$$

The reported t statistic is

$$\frac{2.40}{0.60} = 4.$$

3.14 Practice at Home

1. Starting from

$$\hat{\beta} = \beta + (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \varepsilon,$$

derive

$$\mathbb{E}\left(\hat{\beta}\right) = \beta$$

and

$$\mathrm{Var}\left(\hat{\beta}\right) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}.$$

2. Explain why normality is not required for OLS to be unbiased but is useful for exact finite sample inference.
3. Explain geometrically why the residual degrees of freedom are

$$n - p.$$

4. Explain why

$$\hat{\beta} \quad \text{and} \quad \text{SSE}$$

are independent under the normal linear model.

5. Derive the t statistic for testing

$$H_0 : \beta_j = \beta_{j,0}.$$

6. Explain the connection between a two sided hypothesis test and the corresponding confidence interval.
7. Explain the difference between a confidence interval for

$$\mathbf{x}_0^\top \beta$$

and a prediction interval for a future observation at $\mathbf{x}_0$.

8. Fit a simple regression model in R and identify:

- the coefficient estimates;
- their standard errors;
- the t statistics;
- the p values;
- the residual standard error;
- the residual degrees of freedom.

Summary

This chapter moved from the **geometry of least squares** to the **probability of least squares**.
Under the first two moment assumptions,

$$\mathbb{E}\left(\hat{\beta}\right) = \beta$$

and

$$\mathrm{Var}\left(\hat{\beta}\right) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.$$

Under the additional normality assumption,

$$\boxed{\hat{\beta} \sim N_p\left(\beta, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\right).}$$

The residual sum of squares satisfies

$$\boxed{\frac{\mathrm{SSE}}{\sigma^2} \sim \chi^2_{n-p}.}$$

This leads to the unbiased estimator

$$\boxed{\hat{\sigma}^2 = \frac{\mathrm{SSE}}{n-p}.}$$

The independence of

$$\hat{\beta}$$

and

$$\mathrm{SSE}$$

allows us to replace the unknown σ with $\hat{\sigma}$ and obtain exact t inference.
The same framework extends naturally to:

- linear combinations of regression coefficients;
- general linear hypotheses;

- confidence intervals for the mean response;
- prediction intervals for future observations.

In the next chapter, we will use these results to develop the ANOVA decomposition, the overall F test, and comparisons of nested regression models.

Side Fact: Distribution Results Used in This Chapter

If

$$\mathbf{Z} \sim N_n\left(\mathbf{0}, \mathbf{I}_n\right)$$

and $\mathbf{A}$ is symmetric and idempotent with

$$\operatorname{rank}(\mathbf{A}) = r,$$

then

$$\boxed{\mathbf{Z}^\top \mathbf{A} \mathbf{Z} \sim \chi^2_r.}$$

If

$$Z \sim N(0,1), \qquad U \sim \chi^2_\nu,$$

and Z and U are independent, then

$$\boxed{\frac{Z}{\sqrt{U/\nu}} \sim t_\nu.}$$

If

$$U_1 \sim \chi^2_r, \qquad U_2 \sim \chi^2_\nu,$$

and U_1 and U_2 are independent, then

$$\boxed{\frac{U_1/r}{U_2/\nu} \sim F_{r,\nu}.}$$

These results provide the distributional foundation for regression inference.

References

Montgomery, Douglas C. 2017. *Design and Analysis of Experiments*. John wiley & Sons.

Montgomery, Douglas C, Elizabeth A Peck, and G Geoffrey Vining. 2021. *Introduction to Linear Regression Analysis*. John Wiley & Sons.

Petersen, Kaare Brandt, and Michael Syskind Pedersen. 2008. *The Matrix Cookbook*. Technical University of Denmark.

Seber, George AF, and Alan J Lee. 2003. *Linear Regression Analysis*. John Wiley & Sons.

Part I

Appendix

Matrix Algebra Review

Matrix dimensions

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times p$, then $\mathbf{AB}$ is defined and is $m \times p$.

Transpose rules

For conformable matrices,

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top.$$

Inverse rule

If $\mathbf{A}$ and $\mathbf{B}$ are invertible, then

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}.$$

Symmetric matrices

A matrix $\mathbf{A}$ is symmetric if

$$\mathbf{A}^\top = \mathbf{A}.$$

Covariance matrices are always symmetric.

A good reference of the linear algebra can be found from Petersen and Pedersen (2008).

- Petersen, Kaare Brandt, and Michael Syskind Pedersen. (2008). *The Matrix Cookbook.* Technical University of Denmark.

Part II

Assessments

Assignment 1

This assignment focuses on the foundations of the linear model:

- matrix representation,
- least squares estimation,
- projection
- basic inference.

> **Instructions**
>
> - Submit both your source file (`.qmd` or `.Rmd`) and the rendered HTML or PDF file.
> - Name your source file as `A1_LastName_FirstName.qmd` or `A1_LastName_FirstName.Rmd`.
> - Clearly label every question and subpart.
> - Type all mathematical work using LaTeX.
> - Include all R code used to obtain your answers.
> - Your document must render from beginning to end without error.
> - Do not submit screenshots of code, R output, or mathematical derivations.
> - Round numerical answers to three decimal places when appropriate.
> - For True/False questions, enter only `T` or `F`.
> - For short-answer questions, one or two clear sentences are sufficient unless a derivation is requested.

The total value of the assignment is [**40**].

Question 1: Matrix Representation of the Linear Model [8]

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \dots, 5,$$

with the following observed data:

i	x_i	Y_i
1	0	2
2	1	4
3	2	5
4	3	8
5	4	9

(a) [2]

Which of the following is the correct design matrix $\mathbf{X}$?

☐ **A.**

$$\begin{pmatrix} 0 & 2 \\ 1 & 4 \\ 2 & 5 \\ 3 & 8 \\ 4 & 9 \end{pmatrix}$$

☐ **B.**

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}$$

☐ **C.**

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix}$$

☐ **D.**

$$\begin{pmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 5 \\ 1 & 8 \\ 1 & 9 \end{pmatrix}$$

(b) [2]

Fill in the following table.

Quantity	Dimension / Rank
$\mathbf{Y}$	___
$\mathbf{X}$	___
β	___
rank($\mathbf{X}$)	___

(c) [2]

Which pair is correct?

☐ **A.**

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 5 & 10 \\ 10 & 30 \end{pmatrix}, \qquad \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 28 \\ 74 \end{pmatrix}$$

☐ **B.**

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 5 & 5 \\ 5 & 30 \end{pmatrix}, \qquad \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 28 \\ 74 \end{pmatrix}$$

☐ **C.**

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 30 & 10 \\ 10 & 5 \end{pmatrix}, \qquad \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 77 \\ 28 \end{pmatrix}$$

☐ **D.**

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 5 & 10 \\ 10 & 20 \end{pmatrix}, \qquad \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 28 \\ 77 \end{pmatrix}$$

(d) [2]

Assume

$$\mathbb{E}[\varepsilon] = \mathbf{0}, \qquad \mathrm{Var}(\varepsilon) = \sigma^2 \mathbf{I}_5.$$

For each statement, enter `T` or `F`.

Statement	T/F
$\mathbb{E}[\mathbf{Y}] = \mathbf{X}\beta$	___
$\mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_5$	___
$\mathbb{E}[\mathbf{Y}] = \mathbf{0}$	___
The five errors all have variance σ^2	___

Question 2: Least Squares and Projection [12]

Continue using the data and design matrix from Question 1.

(a) [4]

Starting from

$$S(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta),$$

derive the normal equations

$$\mathbf{X}^\top \mathbf{X}\hat{\beta} = \mathbf{X}^\top \mathbf{Y},$$

and then state the OLS estimator when $\mathbf{X}$ has full column rank.

Keep your derivation concise.

(b) [3]

Using the quantities from Question 1, compute

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix}.$$

Fill in the blanks:

$$\hat{\beta}_0 = \underline{\hspace{3cm}}, \qquad \hat{\beta}_1 = \underline{\hspace{3cm}}.$$

Hence,

$$\hat{Y} = \underline{\hspace{3cm}} + \underline{\hspace{3cm}}x.$$

(c) [2]

For the least squares residual vector

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}},$$

which statement must be true?

- ☐ **A.** $\mathbf{X}\mathbf{e} = \mathbf{0}$
- ☐ **B.** $\mathbf{X}^\top\mathbf{e} = \mathbf{0}$
- ☐ **C.** $\mathbf{e}^\top\mathbf{e} = 0$
- ☐ **D.** $\mathbf{H}\mathbf{e} = \mathbf{e}$

In one sentence, explain the geometric meaning of your choice.

Your explanation:

(d) [3]

For the hat matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top,$$

enter `T` or `F`.

Statement	T/F
$\mathbf{H}^\top = \mathbf{H}$	___
$\mathbf{H}^2 = \mathbf{H}$	___
$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$	___
$\mathbf{H}$ is an orthogonal projection matrix	___

Question 3: Consequences and Geometry of Least Squares [12]

Continue using the data and fitted regression model from Questions 1 and 2.

Recall that

$$\hat{\beta}_0 = 2.0, \qquad \hat{\beta}_1 = 1.8.$$

(a) Fitted Values and Residuals [3]

Compute the fitted values

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}$$

and the residual vector

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}.$$

Fill in the blanks:

$$\hat{\mathbf{Y}} = \begin{pmatrix} \_\_\_\_\_ \\ \_\_\_\_\_ \\ \_\_\_\_\_ \\ \_\_\_\_\_ \\ \_\_\_\_\_ \end{pmatrix}, \qquad \mathbf{e} = \begin{pmatrix} \_\_\_\_\_ \\ \_\_\_\_\_ \\ \_\_\_\_\_ \\ \_\_\_\_\_ \\ \_\_\_\_\_ \end{pmatrix}.$$

(b) Consequences of Including an Intercept [3]

For each statement, enter `T` or `F`.

Statement	T/F
The residuals satisfy $\sum_{i=1}^{n} e_i = 0$.	___
The fitted values satisfy $\bar{\hat{Y}} = \bar{Y}$.	___
The vector $\mathbf{1}_n$ belongs to $\mathcal{C}(\mathbf{X})$.	___
These properties necessarily hold for every regression model without an intercept.	___

(c) Residual-Maker Matrix [3]

Define

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

For each statement, enter `T` or `F`.

Statement	T/F
$\mathbf{e} = \mathbf{MY}$	___
$\mathbf{M}^\top = \mathbf{M}$	___
$\mathbf{M}^2 = \mathbf{M}$	___
$\mathbf{HM} = \mathbf{0}$	___

(d) Rank and Uniqueness [3]

Suppose now that a design matrix $\mathbf{X}$ does **not** have full column rank.

For each statement, enter `T` or `F`.

Statement	T/F
$(\mathbf{X}^\top\mathbf{X})^{-1}$ necessarily exists.	___
The least squares coefficient vector $\hat{\beta}$ may not be unique.	___
The fitted vector $\hat{\mathbf{Y}}$ is still unique.	___
The residual vector $\mathbf{e}$ is still unique.	___

Question 4: Linear Regression in R [8]

For this question, use the built-in R dataset `cars`.

The variables are:

- `speed`: speed of the car in miles per hour;
- `dist`: stopping distance in feet.

Consider the model

$$\text{dist}_i = \beta_0 + \beta_1 \text{speed}_i + \varepsilon_i.$$

Load the data using

```r
data(cars)
```

(a) [4]

Fit the model using `lm()` and complete the table.

```r
fit <- lm(dist ~ speed, data = cars)
summary(fit)
```

```

Call:
lm(formula = dist ~ speed, data = cars)

Residuals:
    Min      1Q  Median      3Q     Max 
-29.069  -9.525  -2.272   9.215  43.201 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -17.5791     6.7584  -2.601   0.0123 *  
speed         3.9324     0.4155   9.464 1.49e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.38 on 48 degrees of freedom
Multiple R-squared:  0.6511,    Adjusted R-squared:  0.6438 
F-statistic: 89.57 on 1 and 48 DF,  p-value: 1.49e-12
```

Quantity	Your answer
$\widehat{\beta}_0$	---
$\widehat{\beta}_1$	---

Which interpretation of $\widehat{\beta}_1$ is correct?

- ☐ **A.** For each 1 mph increase in speed, the stopping distance increases by exactly 3.9324 feet for every car.
- ☐ **B.** For each 1 mph increase in speed, the fitted mean stopping distance increases by approximately 3.9324 feet.
- ☐ **C.** The average speed increases by 3.9324 mph for every extra foot of stopping distance.
- ☐ **D.** A car travelling at 0 mph has a stopping distance of 3.9324 feet.

(b) [4]

Construct $\mathbf{Y}$ and $\mathbf{X}$ manually and reproduce the OLS coefficients using matrix operations.

Complete the following code:

```r
Y <- matrix(cars$dist, ncol = 1)

X <- cbind(
  1,
  cars$speed
)

# Replace NULL with your matrix expression
beta_hat <- NULL

beta_hat
```

```
NULL
```

Then report

$$\widehat{\beta} = \begin{pmatrix} \underline{\hspace{4cm}} \\ \underline{\hspace{4cm}} \end{pmatrix}.$$

Grading Summary

Question	Topic	Value
Question 1	Matrix representation	8
Question 2	Least squares and projection	12
Question 3	Distribution and inference	12
Question 4	Regression in R	8
Total		**40**

Assignment 2

Question: Distribution and Inference [12]

Suppose a multiple linear regression model has $n = 20$ observations and $p = 3$ regression parameters. Assume

$$\mathbf{Y} \sim N_{20}\left(\mathbf{X}\beta, \sigma^2 \mathbf{I}_{20}\right).$$

Suppose

$$\hat{\beta} = \begin{pmatrix} 2.0 \\ 1.5 \\ -0.5 \end{pmatrix},$$

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \begin{pmatrix} 0.20 & 0 & 0 \\ 0 & 0.05 & 0 \\ 0 & 0 & 0.08 \end{pmatrix},$$

and

$$\text{SSE} = 34.$$

Use

$$t_{0.975,17} = 2.110.$$

(a) [3]

Choose the correct distributions.

For $\hat{\beta}$:

- ☐ **A.** $N_3(\mathbf{0}, \sigma^2 \mathbf{I}_3)$
- ☐ **B.** $N_3(\beta, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$
- ☐ **C.** t_{17}
- ☐ **D.** χ^2_{17}

For SSE/σ^2:

- ☐ **A.** $N(0, 1)$
- ☐ **B.** t_{17}
- ☐ **C.** χ^2_{17}
- ☐ **D.** $F_{3,17}$

(b) [2]

Compute

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n-p}.$$

$$\hat{\sigma}^2 = \underline{\hspace{3cm}}.$$

(c) [4]

For testing

$$H_0 : \beta_1 = 0 \qquad \text{versus} \qquad H_1 : \beta_1 \neq 0,$$

where β_1 is the second entry of β, fill in the following:

$$\text{SE}(\hat{\beta}_1) = \underline{\hspace{3cm}},$$

$$t = \underline{\hspace{3cm}},$$

and choose the decision:

☐ **A.** Reject H_0

☐ **B.** Fail to reject H_0

(d) [3]

Which of the following is the 95% confidence interval for β_1?

☐ **A.** $(0.8329,\ 2.1671)$

☐ **B.** $(1.1838,\ 1.8162)$

☐ **C.** $(-0.6671,\ 0.6671)$

☐ **D.** $(0,\ 3.0000)$